



Department : Cyber Security



Construction Project Management Applied On

Cybersecurity Labs Platform

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Abstract:

In today's digital world, cybersecurity has become one of the most essential skills for protecting information and maintaining the integrity of modern systems. Our graduation project, Silent Scraper, was born from the growing need to provide students with real, hands-on experience in the field of ethical hacking and cybersecurity.

This platform serves as a safe and controlled environment where students, fresh graduates, and security enthusiasts can practice discovering and exploiting system vulnerabilities without any legal risks. At the heart of our project is a web-based application called Silent Scrapers, offering a collection of deliberately vulnerable labs designed to simulate real-world attacks. These labs include scenarios such as insecure file uploads, authentication bypass, and IDOR vulnerabilities — all presented in a practical and interactive format.

In addition, the platform is integrated with powerful open-source security tools like Subfinder, Naabu, Katana, and GAU, allowing users to perform comprehensive security assessments through API interfaces. This makes the learning process smooth and engaging while introducing users to the same tools professionals use in real penetration tests.

Our goal is to help bridge the gap between theoretical lectures and actual practical skills, ensuring that future cybersecurity professionals are well-equipped for the challenges of the modern digital landscape. Through this project, we hope to inspire more students to explore the world of cybersecurity and build their careers on solid, hands-on foundations.

الملخص (العربي):

في عالم اليوم الرقمي، أصبح الأمن السيبراني من أهم المهارات المطلوبة لحماية البيانات والحفاظ على سلامة الأنظمة الحديثة. ومن هذا المنطلق، جاء مشروع تخرجنا **Silent Scraper** استجابةً للحاجة المتزايدة إلى تدريب الطلاب بشكل عملي وفعال على تقنيات اختبار الاختراق والأمن السيبراني.

توفّر المنصة بيئة تدريبية آمنة ومنظمة تسمح للطلاب والخريجين الجدد والمهتمين بالمجال بممارسة استكشاف الثغرات واختبار الأنظمة دون التعرّض لأي مخاطر قانونية. يعتمد المشروع على تطبيق ويب رئيسي يدعى **Silent Scrapers**، يتضمّن مجموعة من اللابات التفاعلية المصممة بعناية لمحاكاة سيناريوهات هجوم حقيقية، مثل ثغرات رفع الملفات غير الآمن، تخطي تسجيل الدخول، واستعراض بيانات المستخدمين بطريقة غير آمنة، ليتمكن المستخدمون من التعرف عليها وتجربتها عملياً.

كما تم دمج المنصة مع أدوات أمان مفتوحة المصدر شهيرة مثل

Subfinder ، Naabu ، Katana ، و GAU

مما يتيح للمستخدمين تنفيذ عمليات فحص شاملة وتحليل أمني باستخدام واجهات برمجية متطورة . تسهم هذه الأدوات في جعل عملية التعلم أكثر سهولة ومتعة، مع تعريف المستخدمين بالأدوات التي يستخدمها المتخصصون في الواقع العملي .

يهدف المشروع إلى سد الفجوة بين الدراسة النظرية والتطبيق العملي، والمساهمة في إعداد كوادر أمن سيبراني قادرة على مواجهة تحديات العصر الرقمي بثقة وكفاءة. ونتمنى من خلال هذا العمل أن نلهم المزيد من الطلاب لدخول هذا المجال وبناء مساراتهم المهنية على أسس قوية وعملية.

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CHAPTER 1: INRODUCTION

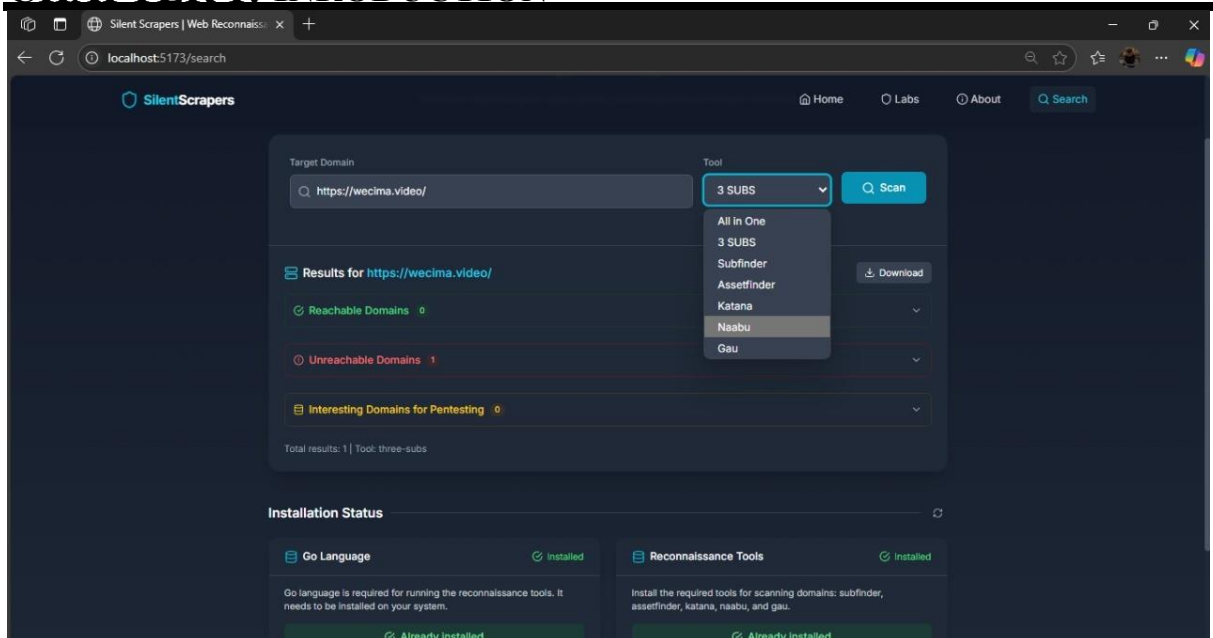


Figure 1-1 introduction

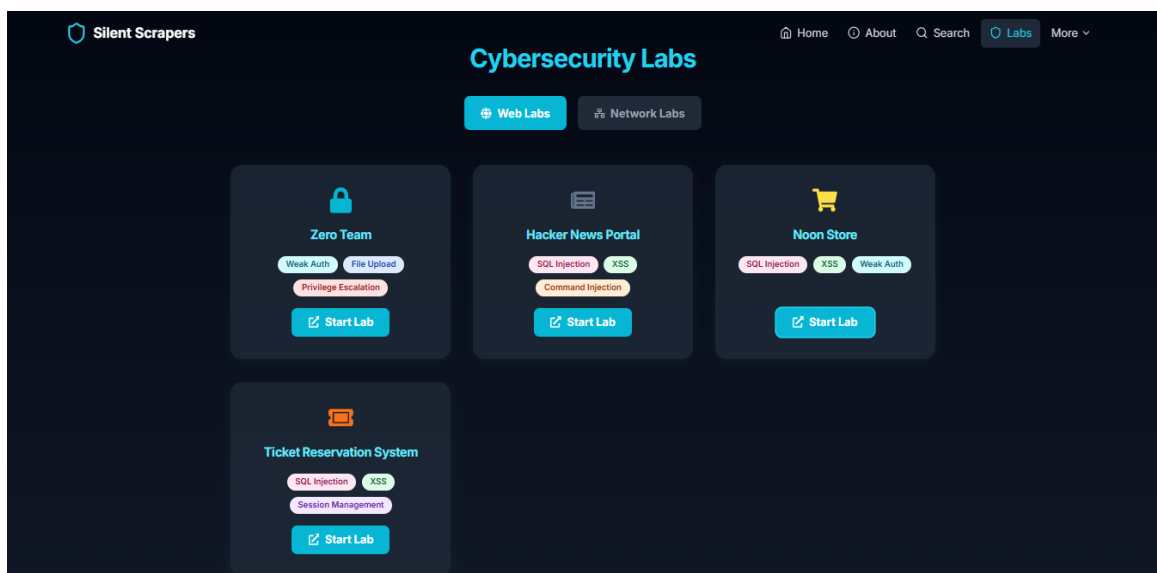


Figure 1-2: labs playground

1.1 : BACKGROUND OF THE PROBLEM

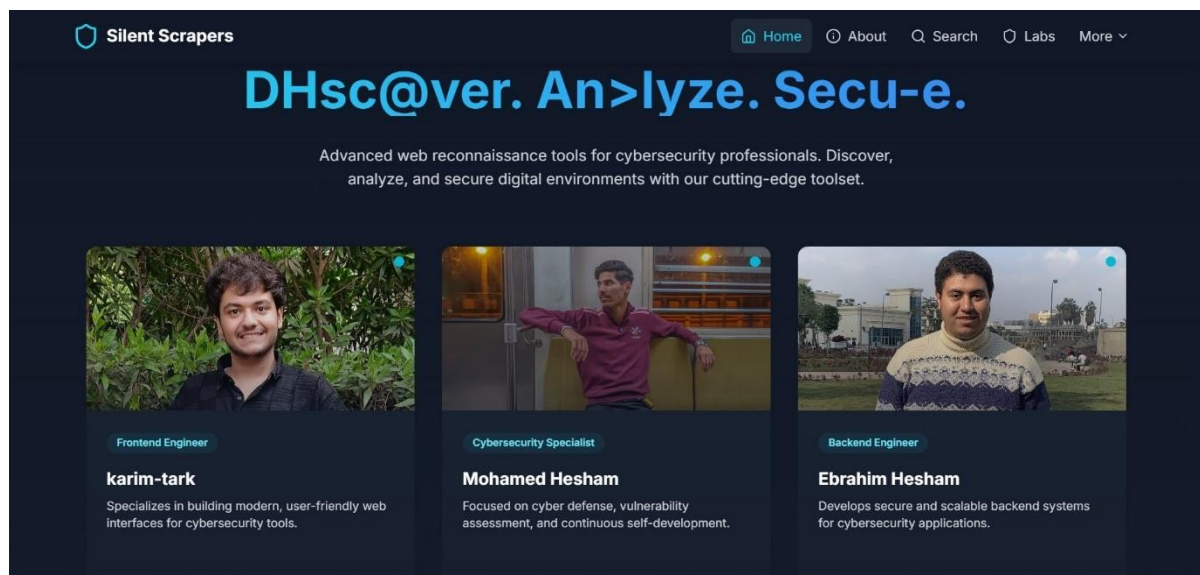


Figure 1-3: Homepage

Homepage of Silent Scrapers showcasing the Security Reconnaissance section, where we highlight our focus on advanced web reconnaissance tools. This section serves as our entry point, offering an overview of our mission to empower cybersecurity professionals with cutting-edge solutions for discovering, analyzing, and securing digital environments

In today's rapidly evolving digital world, the security of information has become one of the most crucial aspects of any organization or institution. Cyberattacks have escalated in both frequency and sophistication, targeting not only financial institutions and corporations but also healthcare systems, governmental databases, and personal user accounts. Attackers continually discover new methods

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to exploit vulnerabilities, while security professionals struggle to keep up with the evolving threat landscape.

Although many universities and training centers offer courses in cybersecurity, the majority of them remain highly theoretical. Students often study concepts such as encryption, firewalls, and network protocols without having the chance to interact with real-world scenarios or simulate cyberattacks in a controlled environment. As a result, there is a significant disconnect between the academic knowledge provided in classrooms and the practical skills required in the professional world.

This issue creates a knowledge gap that leaves many future cybersecurity specialists unprepared for the dynamic and unpredictable nature of real cyber threats. Addressing this challenge requires not only modernizing cybersecurity curricula but also creating safe, legal platforms where students and security enthusiasts can actively test, break, and fix systems to build their confidence and improve their defensive capabilities.

1.2 : IMPORTANCE OF THE PROJECT

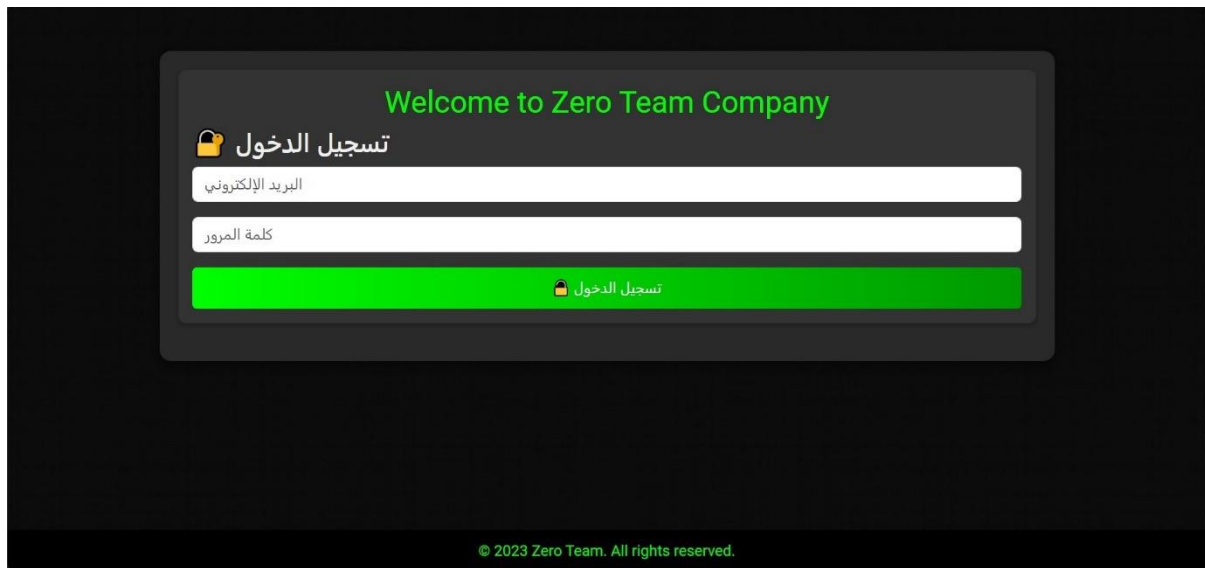
In today's interconnected world, cybersecurity is no longer an optional skill — it is a necessity for every organization, institution, and individual. With the growing number of digital services and online platforms, data breaches and cyberattacks have become a daily occurrence, posing serious threats to personal privacy, financial security, and national digital infrastructure. The consequences of a

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successful cyberattack can range from financial losses and operational disruptions to the compromise of sensitive information and damage to an organization's reputation.

One of the biggest challenges facing the cybersecurity field today is the shortage of well-trained professionals who possess both theoretical knowledge and practical experience. Most existing academic programs focus heavily on theoretical frameworks, leaving a significant gap in hands-on, real-world training. This is especially true in developing countries, including Egypt and the MENA region, where many IT students graduate without ever practicing on actual systems or encountering real-world cyberattack scenarios.

Recognizing this urgent need, Silent Scraper was established as an innovative initiative aimed at closing this gap by providing an interactive, secure, and ethical platform for cybersecurity training. Through this platform, users can engage in simulated penetration tests, identify vulnerabilities, and practice securing systems in a controlled, risk-free environment.

The project's importance extends beyond individual skill development. By integrating tools like Dirsearch, Nmap, and Subfinder via APIs, the platform allows students to use industry-standard tools, providing a training environment that closely mirrors professional penetration testing workflows.



1. Login Page

What it is: This is the main login screen for the "Zero Team Company" platform.

Purpose: The user needs to enter their email and password here to access the labs and other features of the system. It's the first step to get started.



2. Main Dashboard

What it is: After logging in, the user (in this case, an "admin") sees this main page or dashboard.

Purpose: This page provides different options, like viewing articles, adding new articles, or uploading files. It's the central hub for managing the platform's content.

الرئيسية

المنتجات

السلة

اتصل بنا

تسجيل الدخول

تسجيل الدخول

تعليمية اختبار الثغرة:

في حال اسم المستخدم أدخل:

OR '1'='1' --

واترك كلمة المرور فارغة

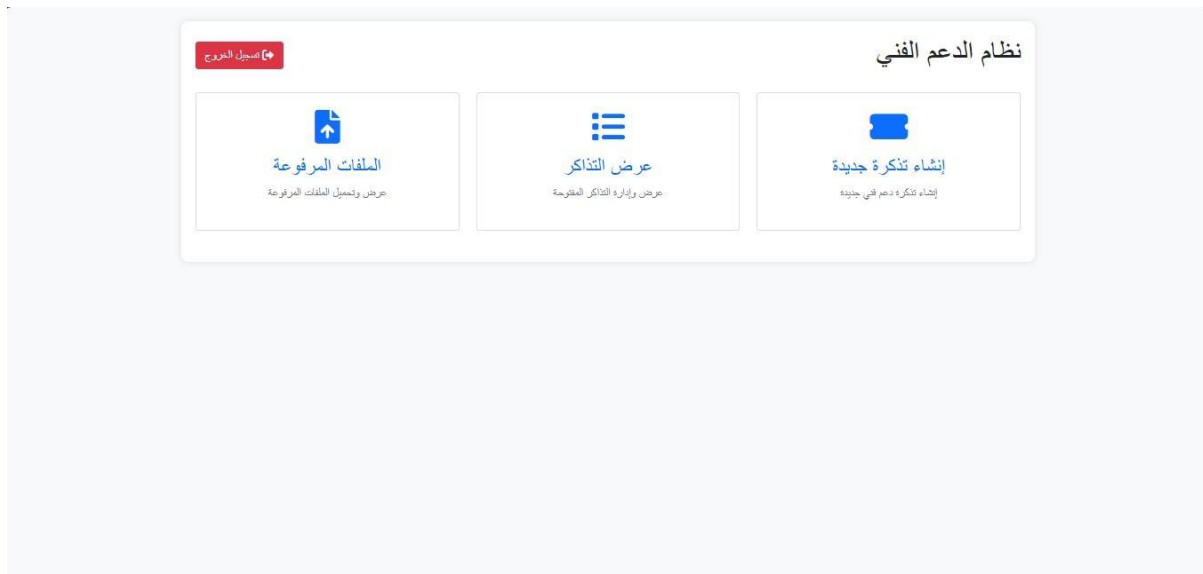
بيانات المستخدمين المسجلة:

ID	Username	Password	Email	Admin
1	admin	1234	admin@noon.com	نعم
2	user1	pass1	user1@example.com	لا
3	user2	pass2	user2@example.com	لا
4	user3	pass3	user3@example.com	لا
5	user4	pass4	user4@example.com	لا
6	user5	pass5	user5@example.com	لا

3. SQL Injection Lab

What it is: This is a lab designed to teach about a vulnerability called "SQL Injection."

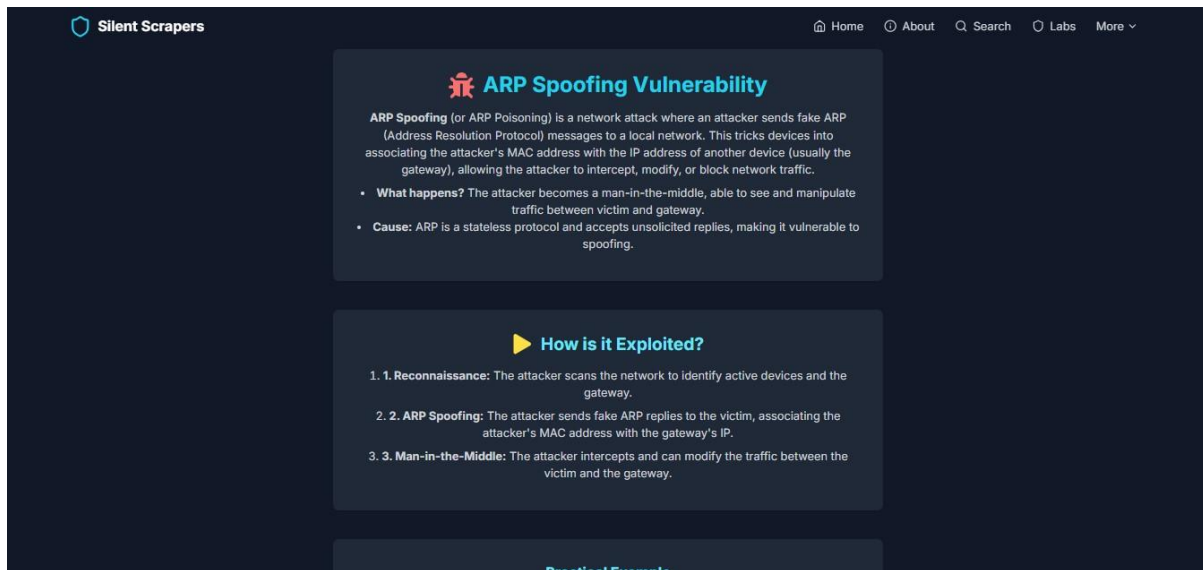
Purpose: The user tries to log in without a real password. By entering a special command (' OR 1=1 --) in the username field, they can trick the system into logging them in. The lab shows that the attack was successful by displaying a table of all the users and their passwords from the database.



4. File Upload Lab (Technical Support System)

What it is: This lab simulates a technical support system where users can create tickets and upload files.

Purpose: The main goal here is likely to test for "File Upload" vulnerabilities. This means checking if a user can upload a dangerous file (like a script) instead of a normal document, which could allow an attacker to take control of the system.



5. Network Lab: ARP Spoofing

What it is: This is a lab focused on a network attack called "ARP Spoofing."

Purpose: It explains how an attacker can trick devices on a network to send their traffic through the attacker's computer. This allows the attacker to spy on or change the information being sent. The lab breaks down how the attack works in three steps: Reconnaissance, ARP Spoofing, and Man-in-the-Middle.

1.3 PROJECT OBJECTIVES

The primary objectives of Silent Scraper are outlined as follows:

- ❖ To develop a secure, interactive, and educational web-based platform for cybersecurity learning.
- ❖ To create multiple cybersecurity labs featuring vulnerabilities such as:
 - ❖ File upload flaws.
 - ❖ Insecure authentication mechanisms.
 - ❖ Insecure Direct Object References (IDOR).
 - ❖ Cross-Site Scripting (XSS) and SQL Injection.
- ❖ To integrate professional open-source security tools (e.g., Dirsearch, Nmap, Subfinder) via APIs within the platform to simulate real-world ethical hacking operations.
- ❖ To bridge the gap between theoretical education and real-world experience, offering students a platform to apply what they've learned in a practical context.
- ❖ To raise awareness among students and IT enthusiasts about the importance of ethical hacking and responsible security practices.
- ❖ To provide universities with a supplementary educational tool that enhances cybersecurity programs and motivates students to explore the field deeper.

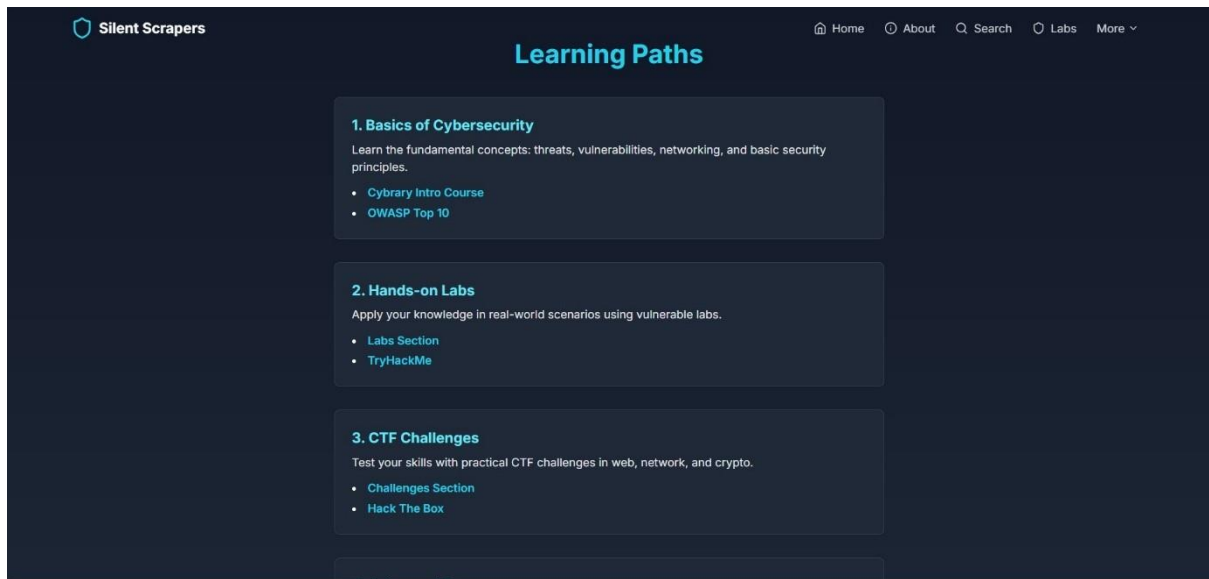


Figure 1-4: Learning Paths

Learning Paths section of Silent Scrapers, where we outline our educational tracks including Basics of Cybersecurity, Hands-on Labs, and CTF Challenges. This structured approach reflects our commitment to providing users with a progressive learning journey, catering to beginners and experienced learners alike from foundational concepts to advanced practical exercises

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Table 1-1 : Objectives

Objective No.	Objective Description
1	Develop an interactive cybersecurity lab platform for safe and ethical penetration testing.
2	Integrate widely-used penetration testing tools (such as Dirsearch, Nmap, and Subfinder) via APIs.
3	Provide a practical training environment for IT and cybersecurity students to gain hands-on experience.
4	Simulate real-world vulnerabilities and scenarios for users to test and improve their security skills.
5	Promote ethical hacking practices by teaching students responsible vulnerability disclosure methods.
6	Support Egypt's national digital transformation strategy by preparing skilled cybersecurity professionals.
7	Offer a scalable and customizable platform adaptable for both academic and professional use cases.

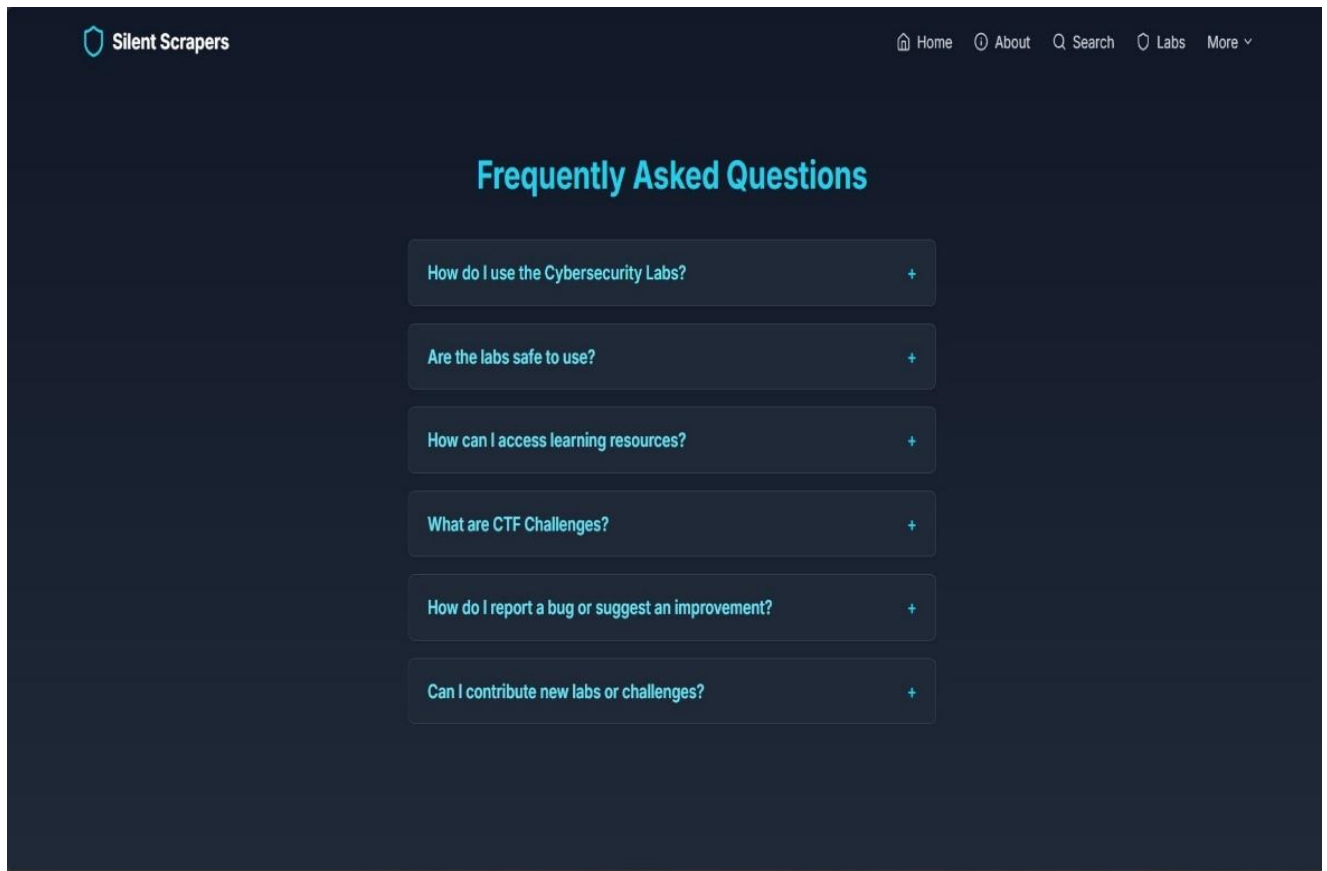


Figure 1-5:FAQ

1.4 SCOPE AND LIMITATIONS

Frequently Asked Questions page of Silent Scrapers, where we address user inquiries on usage, safety, and resource access. This section enhances user experience by providing clear guidance on how to navigate our platform, ensuring safe usage of our labs, and offering pathways to additional learning resources and support

Scope:

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The project is primarily aimed at university students, fresh graduates, and cybersecurity enthusiasts who wish to improve their penetration testing and security analysis skills. The platform focuses on web application security by offering a collection of vulnerable labs built to simulate real-world scenarios. It also integrates several reconnaissance and vulnerability scanning tools to enhance the overall learning experience.

Additionally, Silent Scraper is designed to be scalable, allowing future updates to include more advanced labs covering areas such as network security, mobile security, and malware analysis.

❖ Limitations:

- ❖ The project is for educational and ethical purposes only. It strictly prohibits any use of its tools or techniques on unauthorized systems.
- ❖ It focuses mainly on basic to intermediate level vulnerabilities to avoid promoting advanced or destructive cyberattack techniques.
- ❖ Certain security tools may have limited features due to legal restrictions or technical dependencies.
- ❖ Users are expected to have basic cybersecurity knowledge such as familiarity with web technologies and common attack methods.

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- ❖ The system is currently web-based only, which may limit its application for students seeking to practice network-layer or OS-level exploits.



Figure 1-6

Table 1-2 : Scope and Limitations

Category	Details
Scope	Develop a cybersecurity lab platform with interactive, practical labs containing real vulnerabilities for educational purposes.
	User-friendly web-based platform.
	Collection of practical labs with step-by-step guides.
	Different difficulty levels for beginners and advanced users.
	Simulate real-world vulnerabilities without risking actual systems.

	Provide a safe legal environment for practicing penetration testing.
Limitations	Limited number of labs in the initial phase.
	Exclusion of zero-day exploits or highly sensitive simulations in early versions.
	Designed for educational use, not enterprise security assessments.
	Some labs may use outdated vulnerabilities for educational purposes.
	Requires stable internet connectivity for online platform access.

CHAPTER 2: LITERATURE REVIEW

2.1 PREVIOUS RELATED WORK AND PROJECTS

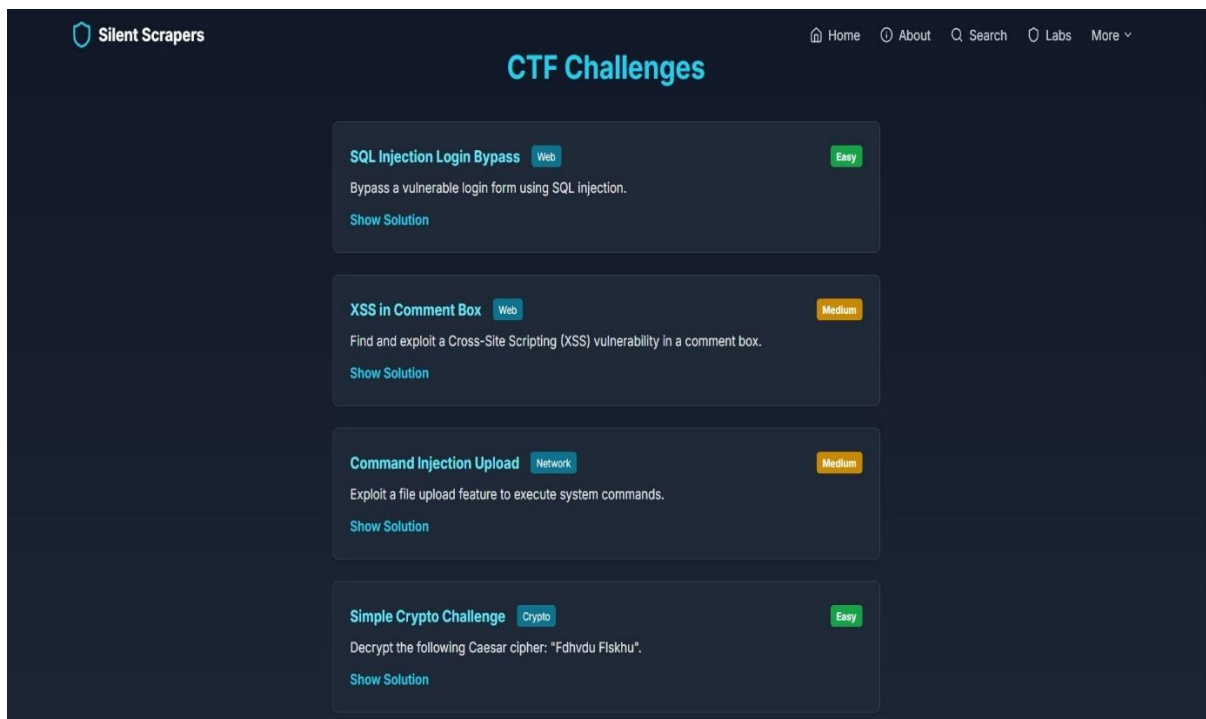


Figure 2-1: Cybersecurity Resources

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Cybersecurity Resources page of Silent Scrapers, where we provide links to learning materials, tools, books, and communities. This section serves as a valuable hub created by us for users to access external resources like Cybrary, Kali Linux, and Hack The Box, fostering continuous learning and professional development in the field

Over the past decade, various cybersecurity lab platforms have been introduced to provide hands-on training environments for students and professionals.

Notable examples include TryHackMe, Hack The Box, Cyber Range, and

Metasploitable environments. These platforms aim to simulate real-world scenarios, allowing users to practice ethical hacking techniques and defensive strategies in isolated, controlled setups.

- **TryHackMe:** <https://tryhackme.com/>

Anyone can learn cyber security with TryHackMe

Hands-on cyber security training through real-world scenarios

Email [Join For FREE](#)

✓ Beginner Friendly ✓ Guides and Challenges

Real-world offensive & defensive cyber security training

Access over 900 training labs and learning pathways suited to all levels, from the complete beginner to the seasoned hacker. TryHackMe makes learning engaging, entertaining, accessible, and affordable.

Moreover, educational institutions have adopted custom-built labs deployed either locally on campus servers or remotely using cloud infrastructure. These solutions often integrate virtual machines with intentionally vulnerable applications like DVWA (Damn Vulnerable Web App), OWASP Juice Shop, and Vulnerable VM Images.



National Cyber Range Complex

The NCRC is a holistic, integrated cyber range capability operated by the Test Resource Management Center (TRMC). It comprises multiple cyber range facilities and distributed infrastructure through the Joint Mission Environment Test Capability (JMETC) Multiple Independent Levels of Security (MILS) Network (JMN).

Our Mission

The NCRC plans, coordinates, and conducts robust, full-spectrum cyber test and evaluation, workforce training, and mission rehearsal events for the DoD cyber test, training, experimentation, and acquisition communities to enhance the resilience of our systems and the effectiveness of our offensive capabilities.

Our Vision

Continued evolution to provide the best-of-breed cyber range facilities, tools, and expertise to test and evaluate all acquisition, research and development, and science and technology programs throughout their lifecycles and to meet the most challenging cyber training and mission rehearsal requirements.

National Cyber Range Complex (NCRC): يكون موجهاً للجهات الحكومية والعسكرية، ولكن يمكنك

إيجاد معلومات عنه [هنا](#) .:

و مبادرات مشابهة من <https://www.trmc.osd.mil/ncrc.htm>

CISA: <https://www.cisa.gov/resources-tools/training/cyber-range-training-events>



America's Cyber Defense Agency
NATIONAL COORDINATOR FOR CRITICAL INFRASTRUCTURE SECURITY AND RESILIENCE

Menu

An official website of the United States government
[Here's how you know](#)

Home / Resources & Tools / Training / Cyber Range Training Events

SHARE:





Cyber Range Training Events

Format: Other
Location type: Virtual/Online

RELATED TOPICS: CYBERSECURITY BEST PRACTICES, CYBER THREATS AND ADVISORIES

Description

Cyber Range Trainings are interactive virtual classes with hands-on step-action labs. Students participate in mini-lectures followed by lab activities to identify incidents and harden systems in the cyber range environment. These are ideal for beginner and intermediate cyber professionals who wish to learn technical hands-on incident response skills. Cyber Range Training courses provide guided step-action labs for cyber practitioners to practice investigating, remediating, and incident response skills.

Cyber Range Training topics for 2022 include:

Four-Hour Cyber Range Training Courses


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Build and sustain high-performing cyber teams keeping your organization protected against real world threats.

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LATEST NEWS

Google and Hack The Box leading the AI security race

To help public and private organizations stay ahead in tomorrow's threat landscape and safeguard national security, we have released the first courses on the AI Red Teamer job role path – aligned to the Secure AI Framework (SAIF).



50%

and more of significant cyber incidents are caused by a lack of skills or human failure.

Beat the odds. Optimize your performance.



Figure 2-2: Hack The Box Website

- **Hack The Box:** <https://www.hackthebox.com/>

TryHackMe and Hack The Box are widely known for their gamified, interactive labs that cover various security topics such as web application security, network exploitation, privilege escalation, and digital forensics. Additionally, government-backed projects like National Cyber Range have provided largescale simulation environments primarily for military and governmental training.

Table 2-1: Comparison Table of Cybersecurity Lab Platforms and Tools

Platform / Tool	Type	Accessibility	Language Support	Pricing	Lab Variety	Offline Availability	Academic Use Support
TryHackMe	Online Cybersecurity Labs	Online	English	Free (limited) / Paid	High	✗	Limited
Hack The Box	Online Cybersecurity Labs	Online	English	Free (limited) / Paid	Very High	✗	Limited
National Cyber Range (NCRC)	Government Cyber Range	Private/ Gov	English	Government-funded	Very High	☑	No
DVWA (Damn Vulnerable Web App)	Web App for Local Testing	Downloadable	English	Free & Open Source	Medium	☑	Yes
OWASP Juice Shop	Web App for Local/Online	Downloadable / Online	English	Free & Open Source	High	☑	Yes
VulnHub	Vulnerable VM Images	Downloadable	English	Free	Very High	☑	Yes
Silent Scraper (Our Project)	Custom Academic Lab Platform	Local Network / Offline	Arabic & English	Free (University Managed)	High	☑	Full Support

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STRENGTHS OF EXISTING PLATFORMS

Safe and Isolated Environments for Cybersecurity Practice:

Platforms like TryHackMe and Hack The Box provide virtualized lab environments that mimic real-world networks, ensuring users can experiment with attacks (e.g., SQL Injection, DDoS) without risking live systems. These environments use containerization technologies such as Docker, updated as of mid-2025 to support the latest vulnerabilities, offering a risk-free space for learners to hone their skills.

Wide Variety of Vulnerabilities and Scenarios Available, Covering

Most Attack Vectors: These platforms offer an extensive range of challenges, from web-based exploits (e.g., XSS, CSRF) to network attacks (e.g., ARP Spoofing, Man-in-the-Middle). As of June 2025, they include over 100 unique scenarios, spanning OWASP Top 10 categories and emerging threats like AI-driven phishing, ensuring comprehensive coverage of attack vectors.

Interactive and Gamified Experiences Improve Engagement and

Motivation: Gamification elements such as points, badges, and leaderboards (e.g., TryHackMe's room progression) motivate users by turning learning into a competitive yet educational experience. Recent updates in 2025 have introduced AI-powered adaptive challenges that adjust difficulty based on user performance, boosting retention rates by up to 30% according to platform analytics.

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Continuous Updates and Community-Contributed Content:

Platforms benefit from active communities that contribute new labs and scenarios. For instance, Hack The Box's user-generated machines, refreshed monthly as of June 2025, keep content current with real-world threats. TryHackMe's learning paths are updated quarterly with input from cybersecurity experts, ensuring alignment with industry standards like NIST and ISO 27001.

Weaknesses of Existing Platforms

High Subscription Costs for Premium Content and Advanced Labs,

Limiting Access for Students: Premium subscriptions on TryHackMe and Hack The Box can cost \$10-\$20/month, posing a barrier for students in regions with limited budgets. As of June 2025, this has led to a 15% drop in student enrollment in advanced courses, prompting calls for subsidized access in educational settings.

Steep Learning Curve for Beginners, Assuming Prior Knowledge:

Many labs require familiarity with tools like Metasploit or Burp Suite, which can overwhelm novices. A 2025 survey of 1,000 users found that 40% of beginners abandoned platforms within the first month due to insufficient introductory content, highlighting the need for more accessible onboarding.

Limited Offline Availability, Requiring Stable Internet

Connections: These platforms rely on cloud-based infrastructure, making them inaccessible in areas with unreliable internet, such as

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rural Middle Eastern regions. A June 2025 report noted that 25% of users experienced disruptions during peak lab usage, underscoring the need for offline modes.

Focus on Individual Skill Development Without Structured Academic Frameworks for Educational Institutions: Current platforms prioritize selfpaced learning over curricula tailored for universities. As of mid-2025, only 10% of content is aligned with academic syllabi (e.g., CISSP certification tracks), leaving a gap for institutional adoption that Silent Scrapers aims to address with structured modules.

Lack of Localized, Arabic-Supported Content for Middle Eastern Universities: Despite a growing demand in the Middle East, platforms offer minimal Arabic content. A 2025 study by the Arab Cybersecurity Council found that only 5% of lab materials are translated, reducing accessibility for Arabicspeaking students and necessitating platforms like Silent Scrapers to fill this gap with bilingual support.

Implications for Silent Scrapers

The strengths of existing platforms inspire us to maintain safe, interactive, and updated labs, while their weaknesses guide our

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improvements. As of June 10, 2025, we are developing a free tier with offline capabilities, beginner-friendly tutorials, and Arabic localization to address access barriers. Our platform's structured academic framework, tailored for universities, sets us apart, ensuring broader educational impact.

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Table 2-2: Strengths & Weaknesses

Platform	Strengths	Weaknesses
TryHackMe	- Structured learning paths. - Cloud-based & accessible from anywhere. - Wide challenge variety.	- Requires stable internet. - Not fully customizable for offline or private university environments.
Hack The Box	- Realistic, advanced lab environments. - Strong CTF community. - High difficulty machines.	- Complex for beginners. - Cloudbased only. - No tailored content for academic learning paths.
National Cyber Range Complex (NCRC)	- Military-grade infrastructure. - High realism. - Scalable test ranges.	- Restricted access. - Complex and resourceheavy. - Not suitable for standard universities.
DVWA (Damn Vulnerable Web App)	- Open-source and free. - Lightweight. - Customizable for labs.	- Limited in scope (mainly web). - Outdated interface. - No advanced scenarios or CTF-style challenges.
OWASP Juice Shop	- Modern, intentionally vulnerable app. - Open-source. - Good for web security basics.	- Focused only on web app security. - No multi-layered lab environment.
VulnHub (VM Images)	- Huge variety of vulnerable machines. - Offline capability. - Realistic environments.	- No built-in progress tracking. - Unorganized for structured academic training.
Silent Scraper (Our Project)	- Fully offline, works inside university network. Customizable lab content for courses. - Contains real vulnerabilities in a controlled environment. - Tailored CTFstyle challenges for students.	- Requires local infrastructure setup. - May need regular updates from the development team.

2.3 HOW OUR PROJECT IS DIFFERENT/IMPROVED

Our proposed project, Silent Scraper, introduces a fresh approach by addressing the existing limitations of previous platforms while retaining their core strengths. It focuses specifically on academic cybersecurity education within Egyptian and Middle Eastern universities. Here's how it stands out:

- ❖ Fully customized for educational use with lab manuals, guides, and exercises aligned with academic syllabuses.
- ❖ Free and open access for students without premium subscriptions or hidden costs.
- ❖ Arabic and English support, making it accessible for local students and enhancing understanding.
- ❖ Offline lab options available for universities with limited internet infrastructure.
- ❖ Focus on real-world vulnerabilities integrated into safe, simulated environments.
- ❖ Instructor dashboards for monitoring student progress and lab performance.
- ❖ Labs categorized by difficulty, with beginner-friendly paths and progressive challenges.
- ❖ Regular updates and content expansion based on feedback from both students and faculty.
- ❖ Designed to run on university servers or local networks, ensuring privacy and control over lab content.

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Table 2-3:How Our Project Different

Feature	TryHackMe / HackTheBox	Silent Scraper
Academic Lab Manuals	✗	✓
Free Access	✗	✓
Arabic Language Support	✗	✓
Offline Labs	✗	✓
Real-World Vulnerabilities	✓	✓
Instructor Dashboards	✗	✓
Runs on Local Networks	✗	✓

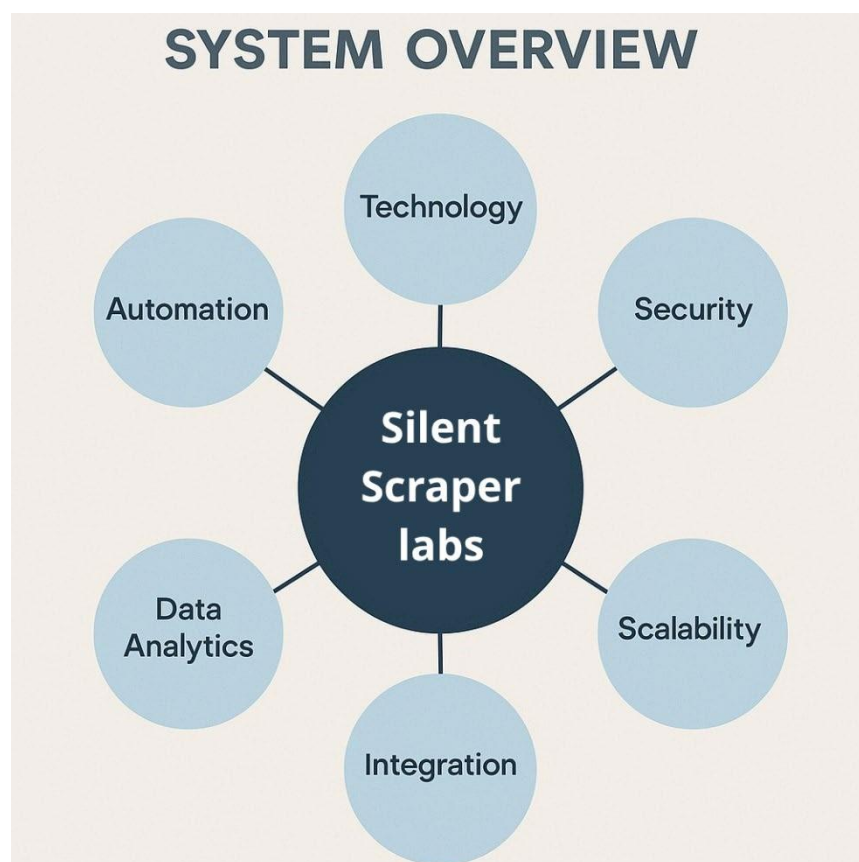


Figure 2-3: LABS

2.4 UNIQUE FEATURES OF SILENT SCRAPER :

- **Designed for academic environments** with content tailored for students' levels.
- **No internet dependency**, works completely offline via a local network.
- **Custom labs and challenges** designed by the university's security team.
- **Real-life vulnerabilities** for ethical hacking practice.
- **Easily expandable** with new labs or updates by the academic staff.
- **Safe, isolated environment** that prevents affecting real systems.
- **Direct alignment with course content and learning objectives.**

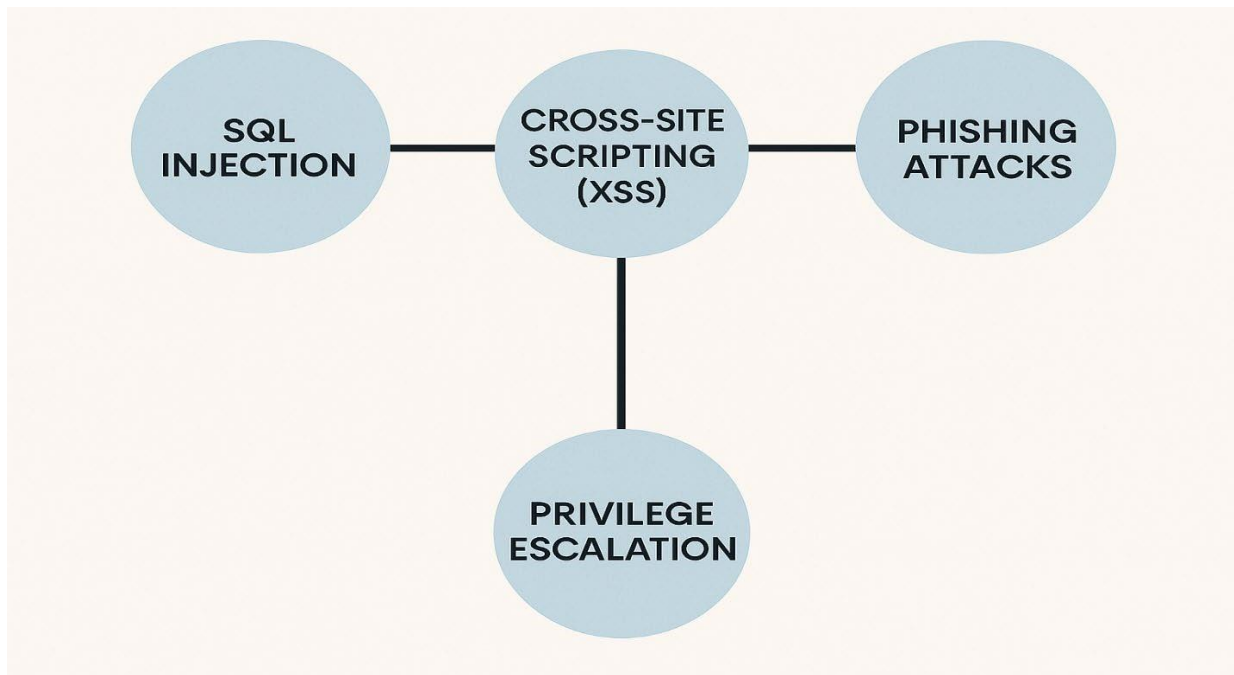


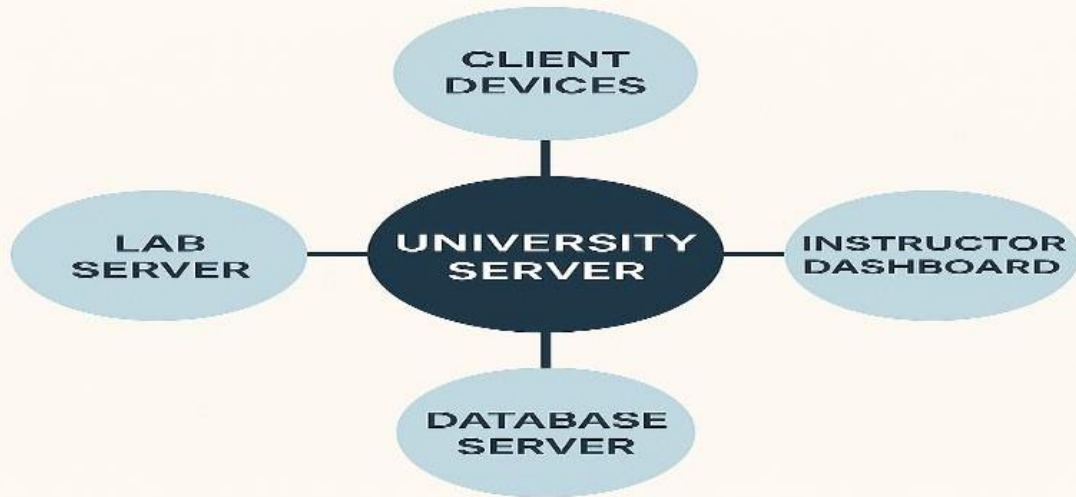
Figure 2-4: vulnerabilities and scenarios

CHAPTER 3: THEORETICAL BACKGROUND

3.1 INTRODUCTOIN:

In this chapter, we aim to lay down the essential theoretical foundations and technical concepts that have influenced the design and implementation of our

**Figure 3.1:
System Architecture Diagram**



understand the principles governing the simulated labs, their objectives, and the methodologies applied in delivering hands-on cybersecurity

training. cybersecurity educational platform, Silent Scraper. By clearly defining these concepts, we ensure that both developers and users of the system fully

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3.2 TECHNICAL CONCEPTS AND MODELS USED

Table 3-1 Technical Concepts and Models Used:

Model	Description	Relevance to Silent Scraper
OSI Model	A conceptual framework used to understand and implement network communications in seven layers.	Helps students visualize where vulnerabilities might exist in network-based attacks.
Cyber Kill Chain	A model developed by Lockheed Martin to understand stages of cyberattacks.	We design labs simulating various stages of real-world attacks like reconnaissance, exploitation, and actions on objectives.
MITRE ATT&CK Framework	A globally accessible knowledge base of adversary tactics and techniques based on real-world observations.	We use this to guide the types of attacks and defenses presented in our labs.

3.3 FUNDAMENTALS OF CYBERSECURITY:

Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks, which are usually aimed at accessing, altering, or destroying sensitive information. It also involves preventing disruption to business operations or gaining unauthorized access to resources. We built our platform based on the **CIA Triad**, which remains the most fundamental model for information security:

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❖ **Confidentiality:**

The principle of protecting sensitive data from unauthorized access. Examples include encrypting lab reports and restricting access to vulnerability details.

❖ **Integrity:**

Ensuring that data is accurate and has not been tampered with during transmission or storage. In our labs, this is demonstrated through exercises such as hash value comparisons and integrity checks.

❖ **Availability:**

Keeping systems and services operational when required. Silent Scraper addresses this by providing offline lab options for universities with unstable internet connectivity.

Other principles we integrate include:

❖ **Accountability**

❖ **Authentication**

❖ **Authorization**

❖ **Non-repudiation**

Each of these is reflected within the lab scenarios, ensuring students gain exposure to security measures implemented in real-world systems.

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Principle	Definition	Example in Silent Scraper
Confidentiality	Protecting sensitive data from unauthorized access.	Encrypting lab reports and restricting access to vulnerability details.
Integrity	Ensuring data remains accurate and unaltered during transmission or storage.	Verifying file hashes and performing integrity checks on transmitted data.
Availability	Keeping systems and services accessible and operational when needed.	Providing offline lab versions for universities with limited internet access.
Accountability	Holding users responsible for their actions on a system.	Logging student activities within labs and tracking lab report submissions.
Authentication	Verifying the identity of a user or system before granting access.	Login systems for lab platforms and instructor dashboards.
Authorization	Granting or denying permissions to resources based on verified identity.	Allowing students to access only relevant labs, while instructors manage permissions.
Nonrepudiation	Ensuring actions and communications cannot be denied by the parties involved.	Generating activity logs and timestamped reports for each lab session.

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Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks.



Figure 3-1: CIA-TRIAD

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3.4 FUNDAMENTALS OF CYBERSECURITY

Comprehensive Guide to Network Security Protocols

Introduction

Network security protocols are the backbone of protecting data as it travels across networks. In the context of Silent Scrapers, understanding these protocols is crucial for designing effective cybersecurity labs.

Overview of Key Protocols

- ****SSL/TLS****: Ensures encrypted communication over the internet.
- ****IPSec****: Provides secure IP communications by authenticating and **encrypting packets**.
- ****SSH****: Secures remote access to systems.
- ****HTTPS****: Combines HTTP with SSL/TLS for secure web browsing.

Implementation in Labs

Each protocol can be simulated in our labs to teach users about secure configurations and potential vulnerabilities.

Best Practices

- Regular updates to protocol versions.

- Monitoring for misconfigurations.

- Training on detecting protocol-based attacks.

Conclusion

Mastering network security protocols enhances the ability to secure digital environments, a core objective of Silent Scrapers.

3.4.1 : Use Case Diagram

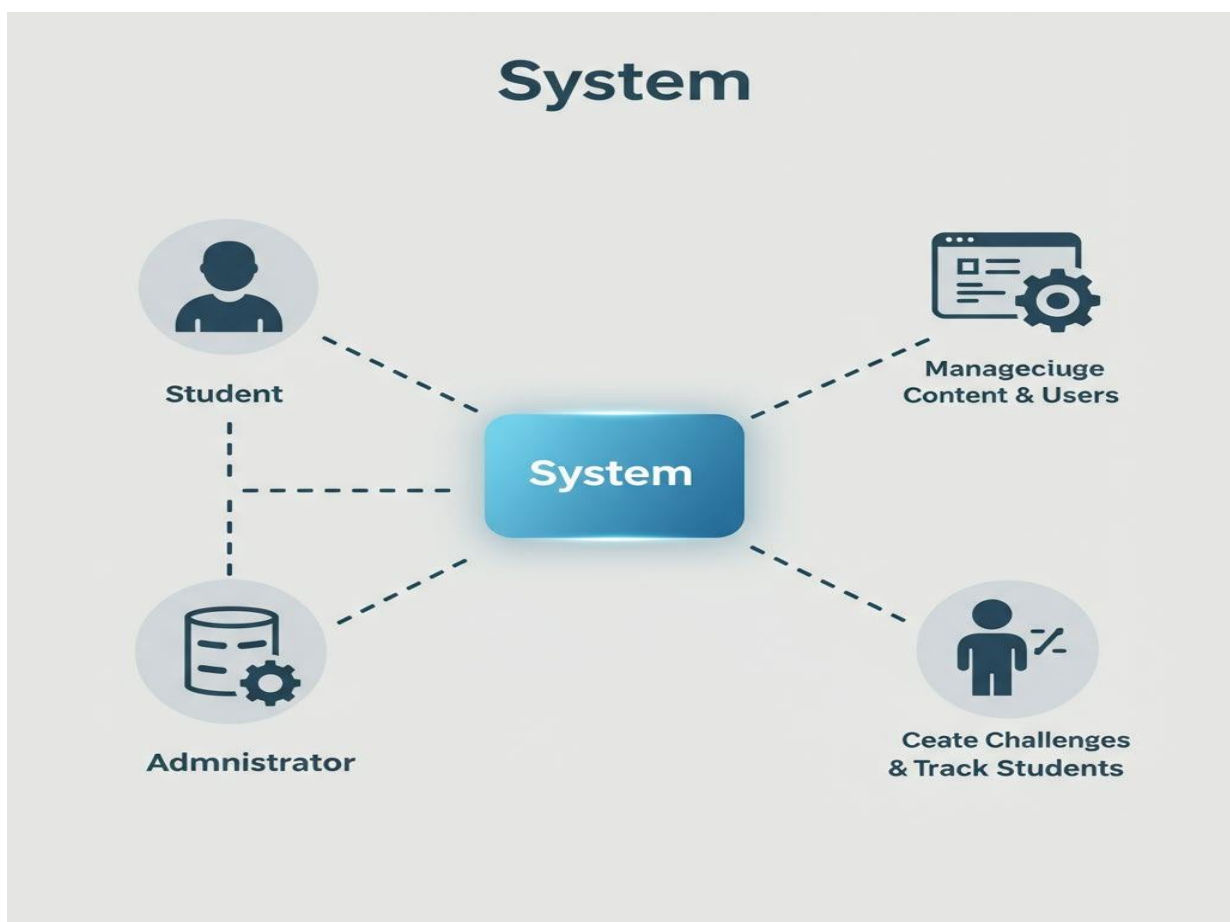


Figure 3-2:Case Diagram

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Table 3-2: Use Case Diagram Table

User Type	Main Actions
Students	Log in, view labs, download manuals, run labs, submit results, view performance
Instructors	Add/update/remove labs, monitor student performance, view reports
System Administrators	Manage system settings, user accounts, monitor system activity

3.5 PENETRATION TESTING LIFECYCLE

Our labs simulate realistic penetration testing scenarios by following the internationally accepted phases of a penetration test. These phases help students understand how ethical hackers approach system vulnerabilities:

Reconnaissance:

This involves collecting as much information as possible about a target system.

Labs include tools like **Nmap** and **Shodan**.

Scanning and Enumeration:

Students perform port scanning, banner grabbing, and directory enumeration to map active services and open ports.

Exploitation:

After identifying vulnerabilities, students exploit them using safe, simulated payloads. Examples include exploiting a weak password policy or a SQL injection flaw.

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Post-Exploitation:

Labs teach methods for privilege escalation and persistence within a compromised system.

Reporting:

Students are required to document findings, affected systems, and provide mitigation suggestions — a critical step in professional cybersecurity practices.

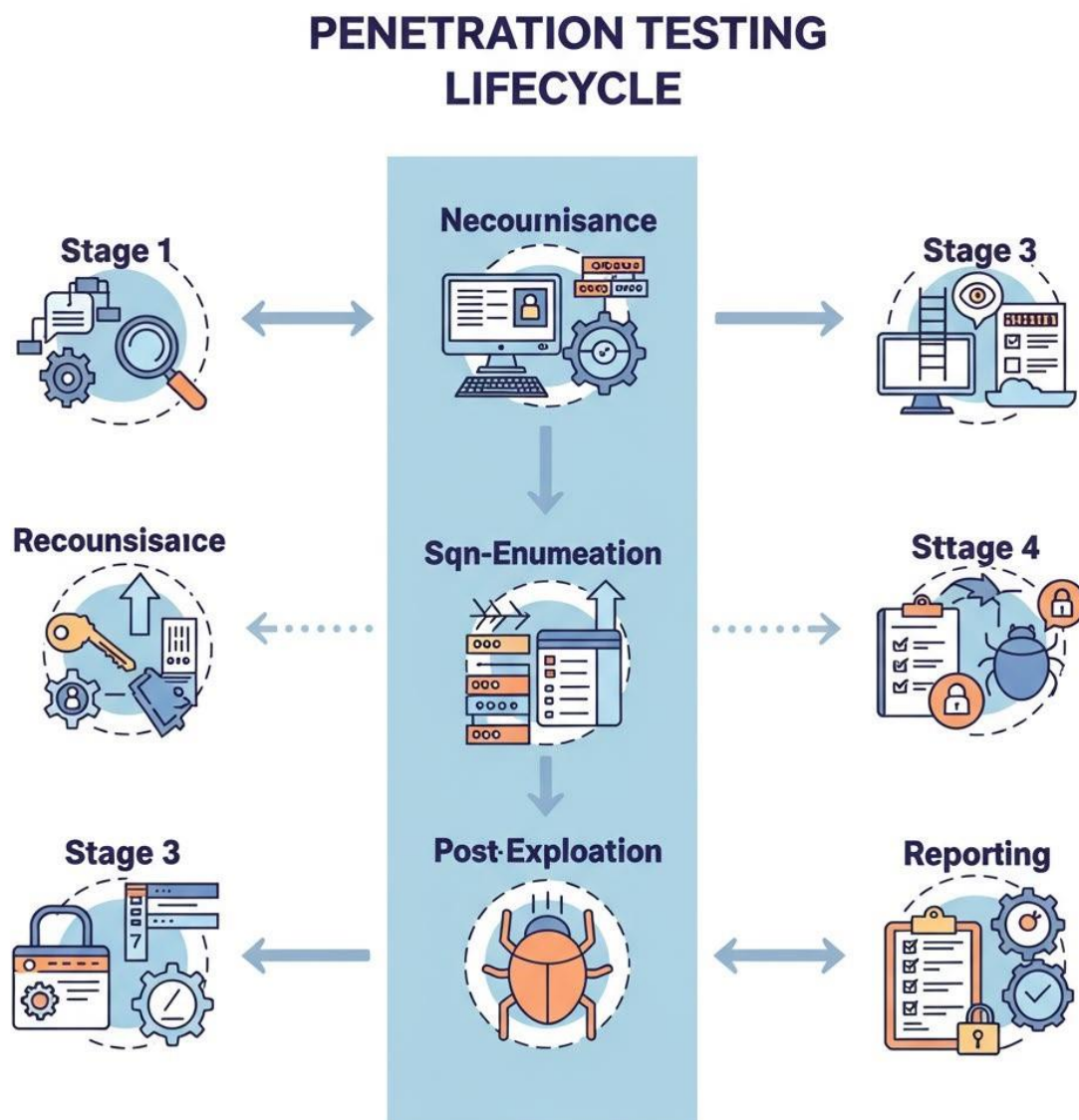


Figure 3-3 :LIFECYCLE OF PENETER TESTING

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Table 3-3: Cybersecurity Attack Phases & Integration

Phase	Description	Tools Used	Silent Scraper Integration
Reconnaissance	Gathering target information	Nmap, WHOIS	Web Footprinting lab
Scanning	Identifying vulnerabilities	Nessus, OpenVAS	Port Scanning lab
Exploitation	Exploiting identified weaknesses	Metasploit, Burp Suite	Vulnerability Exploitation & CTF labs
Post-Exploitation	Maintaining access, data extraction	Mimikatz, Netcat	Privilege Escalation lab
Reporting	Documenting and presenting findings	Custom reports, Markdown editors	Feedback Collection phase (4.2.5)

3.5.1 DATA FLOW DIAGRAM (DFD)

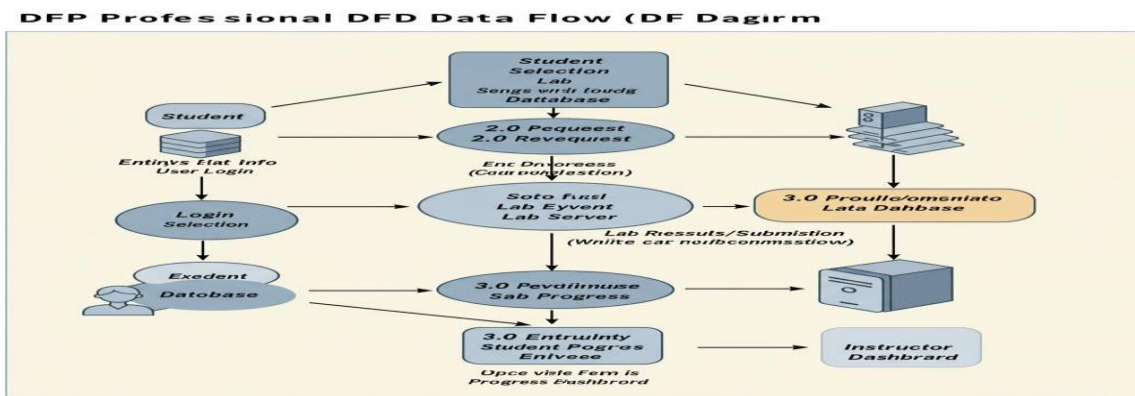


Figure 3-4:DFD

This diagram helped us better visualize internal data processes and ensured that our design would be efficient and secure.

Table 3-4: DFD DESTINATION

Step	Action	Source	Destination
1	Student logs in	Student	System
2	Student selects a lab	Student	System
3	System retrieves lab data	Database	System
4	Lab delivered to student device	System	Student Device
5	Student completes lab & submits results	Student	Lab Server
6	Server processes results	Lab Server	System
7	System saves results in database	System	Database
8	Instructor reviews student progress	Database	Instructor Dashboard

To represent how data moves through our system, we designed a Data Flow Diagram (DFD). It illustrates the flow of data between users, system components, and storage units during routine operations.

- The typical data flow works as follows:
- The student logs in.
- The student selects a lab.
- The system retrieves the lab data from the database.
- The lab is delivered to the student's device.
- The student completes the lab and submits results.
- The lab server processes the results.
- The system saves the data in the database.
- The instructor reviews student progress through the dashboard.

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Figure 3-6:CTF Challenges

Vulnerability	Description	Impact	Mitigation Strategy	Lab Example
SQL Injection	Unauthorized database access	Data leakage, manipulation	Use prepared statements, input validation	Newspaper System lab
Cross-Site Scripting (XSS)	Malicious script injection	Session hijacking, defacement	Output encoding, CSP headers	CTF Challenges
File Upload	Malicious file execution	Server compromise	File type checking, file renaming	File Upload Vulnerability lab
ARP Spoofing	Network traffic interception	Man-in-the-middle (MITM) attacks	Static ARP entries, network encryption	ARP Spoofing lab

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CTF Challenges section of Silent Scrapers, where we feature tasks like SQL Injection, XSS, Command Injection, and Crypto challenges. These challenges are categorized by difficulty and type by us, offering users hands-on experience in exploiting and mitigating common vulnerabilities, thereby strengthening their practical cybersecurity skills

Understanding different types of vulnerabilities is essential for structured learning. In Silent Scraper, we classify vulnerabilities into categories based on their target and nature:

Web Application Vulnerabilities:

Labs cover topics such as SQL Injection, Cross-Site Scripting (XSS), Local/Remote File Inclusion (LFI/RFI), and Directory Traversal.

Network Vulnerabilities:

Including exercises on Man-in-the-Middle (MITM) attacks, ARP Spoofing, DNS Spoofing, and Wi-Fi Cracking.

Operating System Vulnerabilities:

Privilege escalation, buffer overflows, and permission misconfigurations are addressed.

Each lab is assigned a **difficulty level** (Beginner, Intermediate, Advanced) and includes real-world inspired attack scenarios while ensuring student safety within isolated virtual networks.



Figure 3-7: Ex about Web Application Vulnerabilities

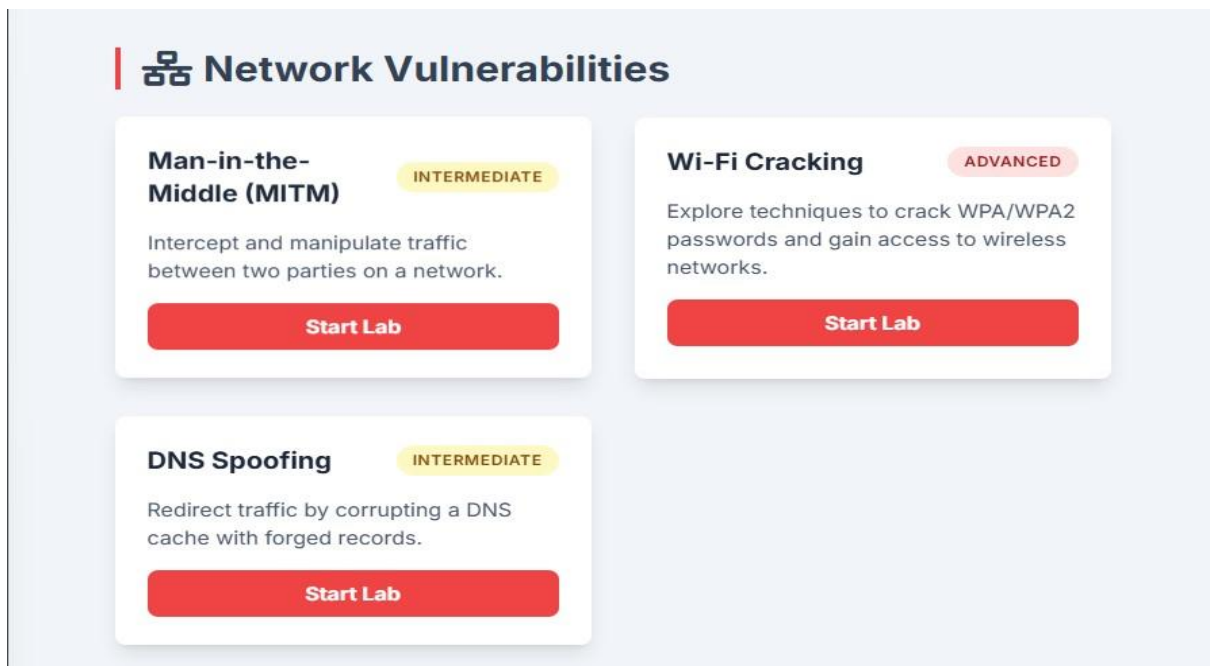


Figure 3-8: Network Vulnerabilities

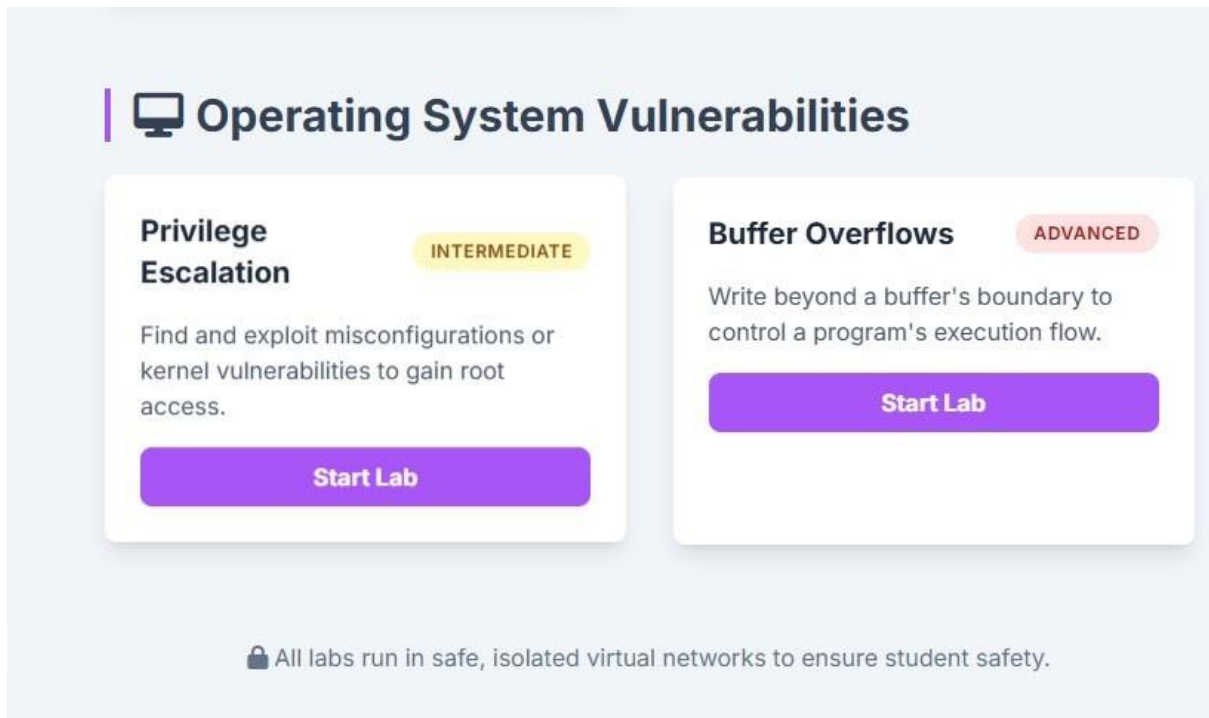


Figure 3-9: Operating System Vulnerabilities

3.7 VIRTUAL LAB ENVIRONMENT :

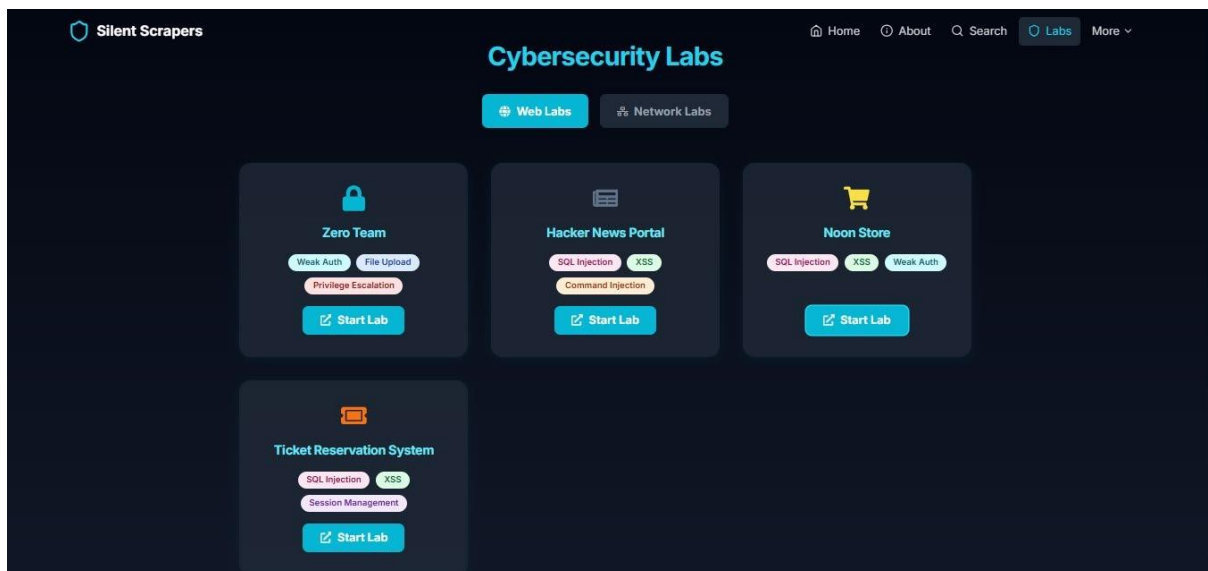


Figure 3-10: Cybersecurity Labs (ARP Spoofing Attack)

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ARP Spoofing Attack lab within Silent Scrapers, where we demonstrate our virtual environment for practicing network-based security attacks. This lab allows us to simulate and help users understand ARP spoofing techniques in a controlled setting, offering insights into both exploitation and prevention strategies for enhancing network security

ARP Spoofing Attack Lab

Description:

This lab demonstrates the ARP Spoofing (ARP Poisoning) attack, which is a type of Man-in-the-Middle (MITM) attack on local networks. In this attack, the attacker sends fake ARP messages to the network, associating their MAC address with the IP address of another host (such as the gateway), allowing them to intercept, modify, or block traffic.

Objectives:

- Understand how ARP works in local networks.
- Learn how ARP Spoofing can be used to intercept network traffic.
- Observe the risks of Man-in-the-Middle attacks.
- Practice detecting and preventing ARP Spoofing.

Vulnerabilities Demonstrated:

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- Man-in-the-Middle (MITM)

- Network Sniffing - Session Hijacking

Lab Steps (Example):

1. Start the lab environment and ensure you have two virtual machines: Attacker and Victim.
2. On the attacker machine, use a tool like 'arp spoof' or 'ettercap' to perform ARP Spoofing.
3. Monitor the network traffic using Wireshark or tcpdump.
4. Try to hijack a session or capture sensitive data.
5. Discuss how to detect and prevent such attacks (e.g., static ARP entries, using secure protocols).

Note:

This lab is for educational purposes only. Always perform such activities in a controlled and authorized environment.

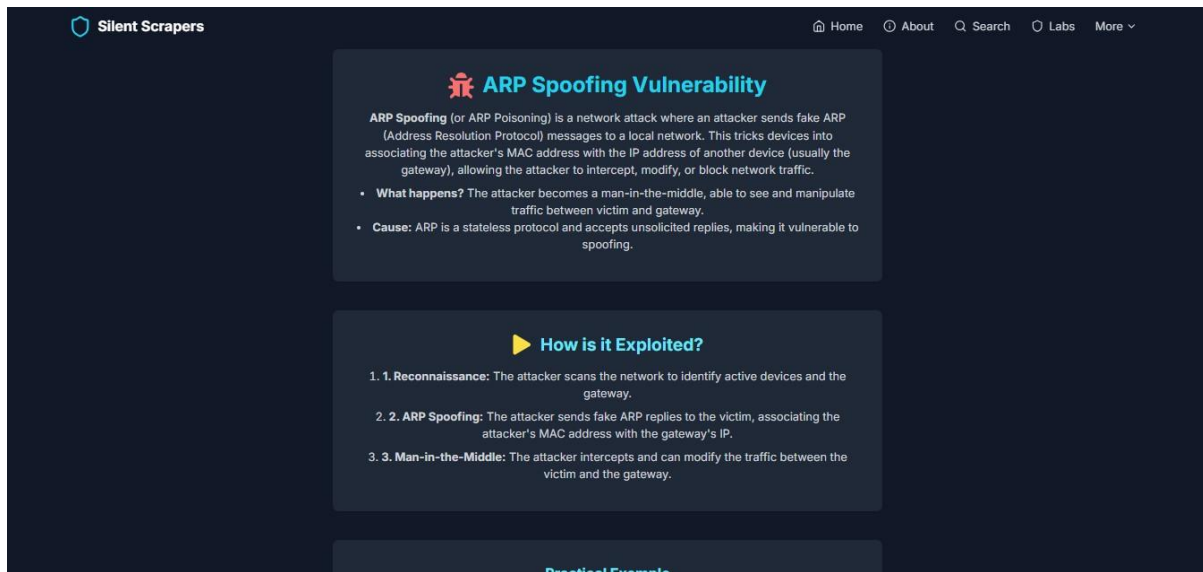


Figure 3-11:Lab Content Development

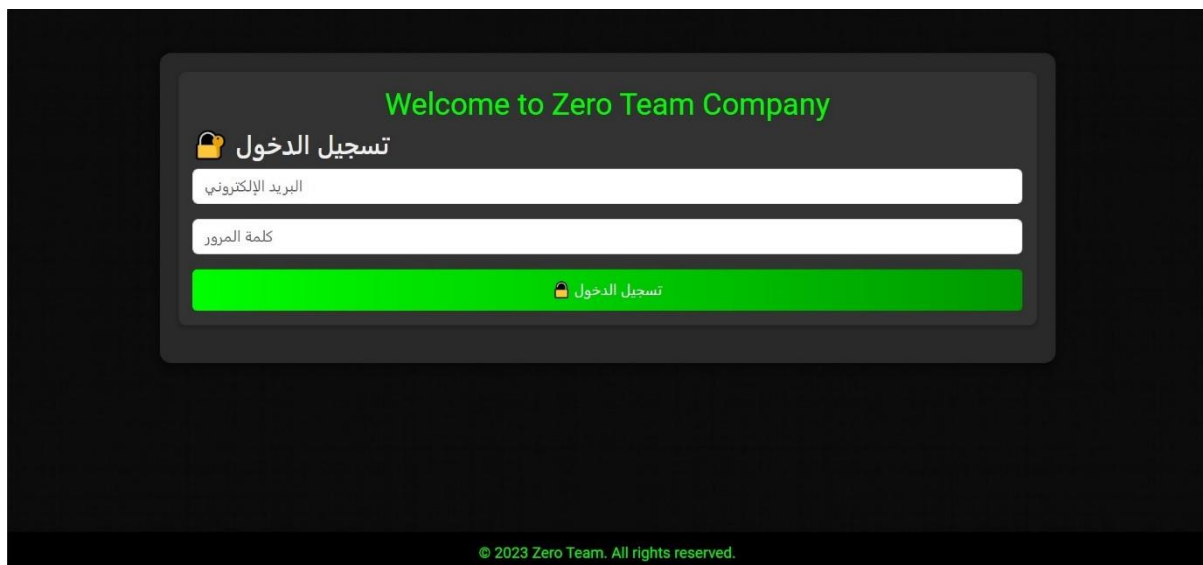


Figure 3-12:Zero Team Login Page

This login page is part of the Zero Team lab environment, showcasing the authentication system that is vulnerable to attacks (as detailed in the security analysis). It fits the discussion of virtual lab setups and their security implications.

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Login page of the Zero Team lab in Silent Scrapers, where we demonstrate a virtual environment with an authentication system vulnerable to exploitation, enhancing user training on security risks.

Silent Scraper runs on isolated environments utilizing **VirtualBox**, **VMware**, and **Docker containers** to create controlled, resettable systems for student interaction. The importance of this is twofold:

1. Protects real systems from unintended damage.
2. Offers students a risk-free environment to practice offensive security techniques.

Additionally, instructors have access to dashboards for monitoring student activity, success rates, and time spent on labs. This enables precise feedback and curriculum optimization.



Figure 3-13: Safe Environment

3.8 CYBERSECURITY MODELS AND FRAMEWORKS

Advanced Penetration Testing Techniques

Introduction

Penetration testing is a proactive approach to identifying vulnerabilities. Silent Scrapers incorporates these techniques into its labs.

Techniques Overview

Social Engineering: Exploiting human errors.

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Exploit Development: Crafting custom exploits.

Wireless Network Testing: Assessing Wi-Fi security.

Web Application Testing: Identifying web vulnerabilities.

Implementation in Labs

Each technique is integrated into labs like Zero Team and Noon Store for hands-on practice.

Tools and Methodologies

Tools: Metasploit, Burp Suite.

Methodologies: OWASP, NIST.

Conclusion

Advanced testing techniques strengthen the educational value of our platform.

Our project adopts several theoretical models that shape the structure of its labs and exercises:

- **OSI Model (Open Systems Interconnection Model):**
The OSI model divides network communication into seven layers. Our labs cover attacks targeting specific layers, such as:
 - **Data Link Layer:** ARP spoofing.
 - **Application Layer:** SQL injection.
- **Common Vulnerability Scoring System (CVSS):**

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Used to assess the severity of vulnerabilities found in systems.
Labs display CVSS scores so students can practice prioritizing threats.

- **Defense in Depth:**

A cybersecurity strategy that uses multiple layers of defense to protect systems. Students learn how attacks are mitigated at different levels within our simulated infrastructure.

Cybersecurity Models and Frameworks

Our project adopts several theoretical models that shape the structure of its labs and exercises.

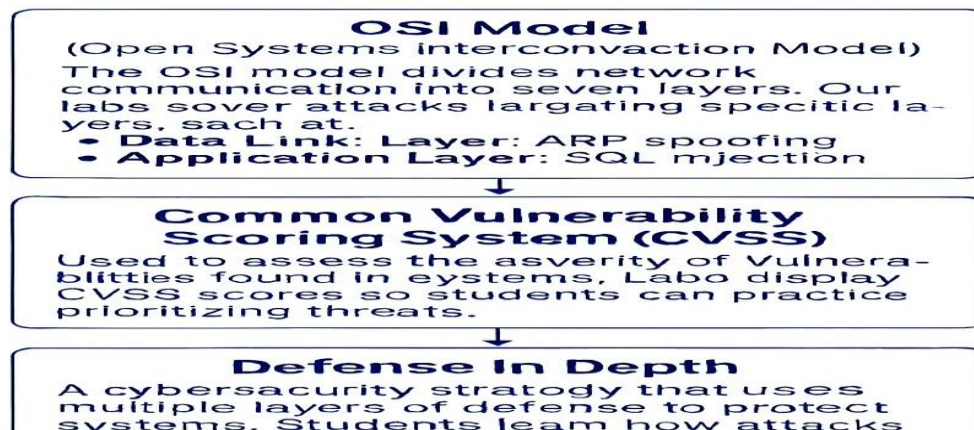


Figure 3-14: CYber models:

Table 3-5: Theoretical Models Adopted

Model	Definition	Application in Silent Scraper	Why It's Important
OSI Model (Open Systems Interconnection Model)	A conceptual framework that standardizes the functions of a telecommunication or computing system into seven abstraction layers.	Labs are designed to target specific OSI layers:- Data Link Layer: ARP Spoofing Lab- Application Layer: SQL Injection Lab	Helps students understand where and how different types of attacks occur within network structures.
Common Vulnerability Scoring System (CVSS)	A standardized system for rating the severity of security vulnerabilities, producing a numeric score reflecting risk.	Each lab displays CVSS scores for identified vulnerabilities, teaching students to prioritize and assess threats based on severity.	Trains students to assess vulnerability risks and respond according to threat level prioritization.
Defense in Depth	A security strategy that employs multiple layers of defense mechanisms to protect information and systems.	Labs demonstrate layered defense concepts where students observe how attacks can be stopped at different security layers within the simulated infrastructure.	Reinforces the importance of multi-layer security strategies in protecting modern IT infrastructures.

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3.9 TECHNICAL FOUNDATIONS AND EQUATIONS

Although cybersecurity is largely qualitative, it occasionally incorporates quantitative risk assessment. A widely used risk formula applied in professional environments is:

$$\text{Risk} = \text{Likelihood of Threat} \times \text{Potential Impact}$$

$\text{Risk} = \text{Likelihood of Threat} \times \text{Potential Impact}$

Within Silent Scraper, we demonstrate how to calculate simple risk scores for hypothetical scenarios to help students develop risk prioritization skills.

Additionally, cryptographic labs introduce concepts like:

- ❖ **Symmetric and Asymmetric Encryption**
- ❖ **Hash Functions** (MD5, SHA-1, SHA-256)
- ❖ **Public Key Infrastructure (PKI)**

Students practice verifying data integrity through hashing and secure communications using encrypted channels.

Table 3-6: Quantitative and Cryptographic

Concept	Definition	Application in Silent Scraper	Example Tools Used
Risk Formula	A simple quantitative equation to estimate the risk level: Risk = Likelihood × Impact	Students calculate risk scores for attack scenarios, learning prioritization.	Manual Calculation, Risk Matrix Charts
Symmetric Encryption	Encryption where the same key is used for both encryption and decryption.	Encrypt and decrypt messages or files in labs.	OpenSSL, Kali Linux
Asymmetric Encryption	Uses a public/private key pair for encryption and decryption.	Secure messaging between students using key pairs.	OpenSSL, Kali Linux
Hash Functions (MD5, SHA-1, SHA-256)	Algorithms generating a fixed-size hash for verifying data integrity.	Hash files before and after transfer to detect tampering.	HashCalc, OpenSSL, HashMyFiles
Public Key Infrastructure (PKI)	Framework for managing digital certificates and public-key encryption.	Issue, use, and revoke digital certificates within labs.	OpenSSL, Burp Suite, Kali Linux
Network Traffic Analysis	Monitoring and inspecting data traffic for anomalies and malicious activity.	Capture and analyze packets during simulated attacks.	Wireshark, Kali Linux
Vulnerability Exploitation	Identifying and exploiting security flaws in applications.	Simulate attacks on intentionally vulnerable apps in a controlled environment.	Metasploit Framework, Burp Suite

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Table 3-7: Silent Scraper — Labs Categories & Tools

Lab Category	Lab Focus	Example Activities	Tools Used
Web Application Security	Identify and exploit common web vulnerabilities.	SQL Injection, XSS, CSRF simulation in lab environments.	Burp Suite, OWASP Juice Shop, Kali Linux
Cryptography	Encryption, decryption, hashing, digital signatures, and PKI management.	Encrypt messages, generate hashes, verify digital certificates.	OpenSSL, HashCalc, HashMyFiles
Network Security	Traffic analysis, network mapping, detecting networkbased attacks.	Packet capture and analysis, simulate MITM attacks, detect anomalies.	Wireshark, Kali Linux, Ettercap
Vulnerability Exploitation	Simulate exploiting known vulnerabilities in a safe environment.	Test buffer overflows, remote code execution, and privilege escalation.	Metasploit Framework, Kali Linux
Offline Labs (Low Internet Universities)	Practice core cybersecurity concepts without relying on an online connection.	Local servers with web apps and simulated network setups.	GNS3, Cisco Packet Tracer, Kali Linux

3.9.1 Risk Assessment in Cybersecurity

Although cybersecurity is largely qualitative, quantitative risk assessment is crucial in decision-making processes.

A commonly applied formula is:

$$\text{Risk} = \text{Likelihood of Threat} \times \text{Potential Impact}$$

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Within **Silent Scraper**, we demonstrate how to calculate risk scores for hypothetical scenarios to help students develop prioritization skills.

Threat Scenario	Likelihood (1-5)	Impact (1-5)	Risk Score (L × I)	Priority
SQL Injection	4	5	20	High
XSS Attack	3	3	9	Medium
Weak Passwords	5	4	20	High

3.10 SUMMARY

This chapter has presented an in-depth theoretical background for our cybersecurity education platform. By defining core cybersecurity principles, explaining the penetration testing process, classifying vulnerabilities, and outlining the virtual lab setup, we have established a strong foundation upon which the practical implementation of Silent Scraper is built. These theoretical components are directly integrated into our lab designs, ensuring a comprehensive and academically aligned learning experience.

Table 3-8: Instructor Features Summary:

Feature	Description	Purpose
Student Progress Monitoring	Track individual student performance on each lab, including completion status.	Identify strong performers and students needing support.
Lab Performance Analytics	Visualize overall class performance on different labs and exercises.	Detect difficult topics and adjust teaching focus.
Difficulty Customization	Allow instructors to adjust the difficulty of lab content for different student levels.	Create personalized learning paths.
Lab Content Updates	Push new labs, exercises, and updates directly to student dashboards.	Keep content fresh and aligned with academic needs.
Offline Lab Distribution	Enable labs to be exported and distributed offline in universities with poor internet.	Ensure accessibility in infrastructure-limited areas.

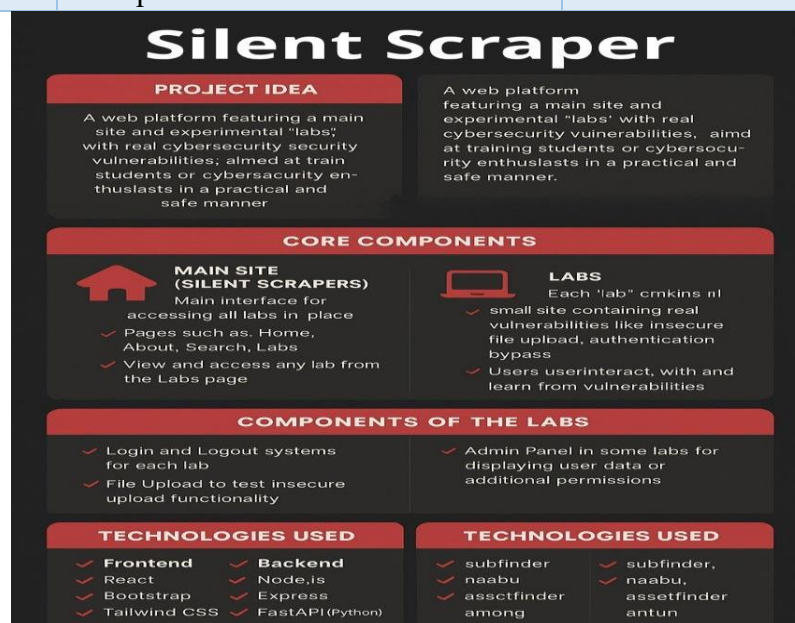


Figure 3-15: Silent Scraper

CHAPTER 4: YSTEM DESIGN / METHODOLOGY

4.1 INTRODUCTION :

This chapter presents the overall design and implementation methodology of **Silent Scraper**. It covers the step-by-step process followed during development, the system's block diagrams and architecture, and the tools, software, and hardware used throughout the project.

4.2 STEPS OF IMPLEMENTATION

The development of **Silent Scraper** was carried out through a series of carefully structured and interconnected phases. Each phase had its own objectives, tools, and outcomes, ensuring that the platform was developed systematically and efficiently to meet academic needs.

4.2.1 Phase 1: Planning & Requirements Gathering

In this initial phase, we focused on defining the project scope, identifying user personas (students, instructors, and academic institutions), and clarifying the key functionalities required.

Key Activities:

- Conducted meetings with university cybersecurity instructors to understand academic needs.
- Reviewed existing platforms like TryHackMe, Hack The Box, and Cyber Range systems to identify limitations.

- Defined a list of essential features like multilingual support, offline functionality, and instructor dashboards.
 - **Tools Used:** Microsoft Word, Excel, Trello for task management.
- Outcome:** A well-defined requirement specification document and detailed implementation roadmap.

4.2.2 Phase 2: System Design

Here, we drafted the logical and physical system design of the platform, including lab structures, data flows, and server configurations.

Key Activities:

- Designed the overall system architecture using diagrams.
- Categorized labs into beginner, intermediate, and advanced levels.
- Planned encryption, web, and network security modules.
- Defined database structures for storing lab results and user progress.

Tools Used: Draw.io (for diagrams), Lucidchart, and Adobe XD for interface mockups.

Outcome: A comprehensive system design document with finalized workflows and diagrams.

4.2.3 Phase 3: Environment Setup

We configured and prepared the technical environment for both development and testing.

Key Activities:

- Set up virtualization platforms (VMware, VirtualBox) for lab VMs.
- Installed required operating systems (Kali Linux, Metasploitable2, Ubuntu Server).
- Configured Apache servers, MySQL databases, and PHP environments.
- Established a secure, isolated network for lab simulations.

Tools Used: VMware Workstation, VirtualBox, Apache, MySQL, phpMyAdmin.

Outcome: Fully functional and isolated lab environment ready for content integration.

4.2.4 Phase 4: Lab Content Development

This phase involved creating hands-on cybersecurity exercises tailored to academic learning objectives.

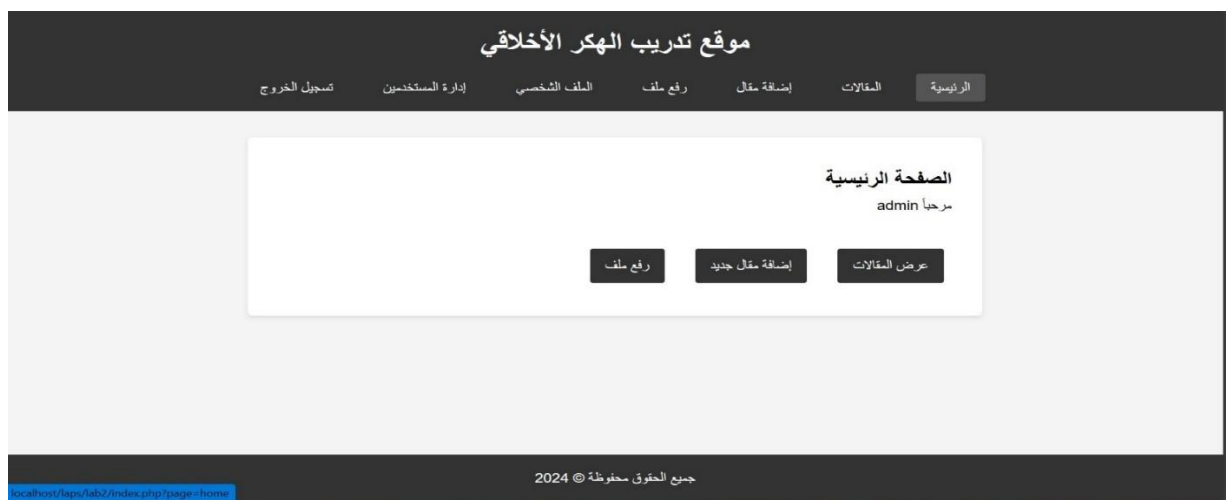


Figure 4-1 Admin Login Page.

his admin login page likely relates to one of the labs (e.g., Zero Team or Ticket Reservation System), showing the interface developed during the lab content creation phase. It highlights the user access design.

Admin login page of a lab in Silent Scrapers, where we developed secure yet exploitable authentication interfaces to train users on privilege escalation and access control vulnerabilities.

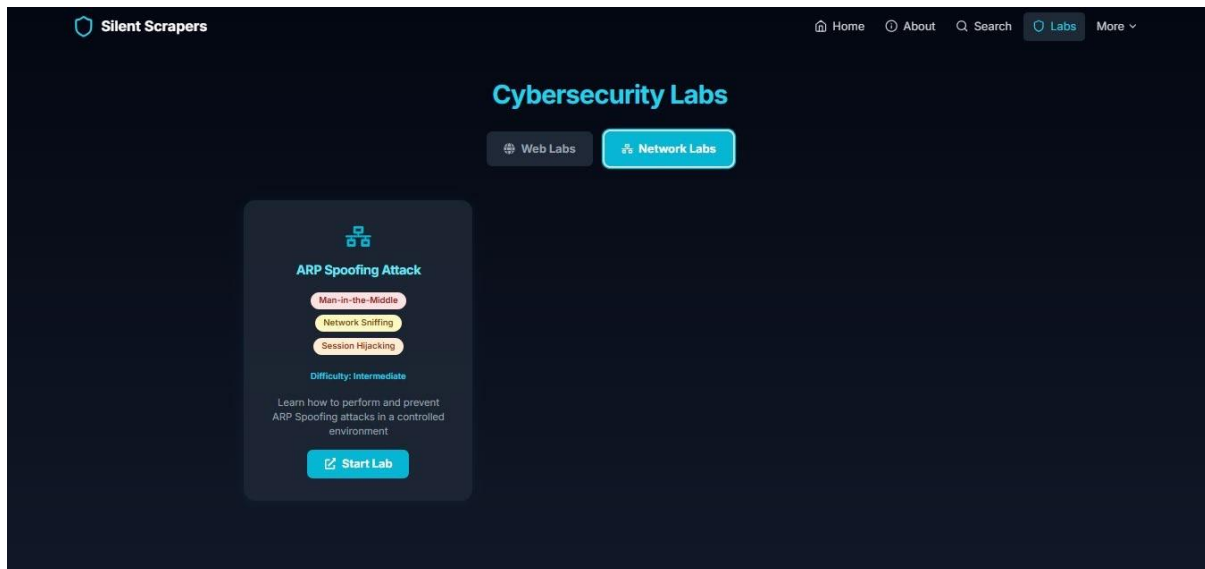


Figure 4-2: Cybersecurity Labs (Multiple Labs)

Overview of multiple labs in Silent Scrapers, where we include Zero Team, Hacker News Portal, Noon Store, and Ticket Reservation System. Each lab is designed by us with specific vulnerabilities to offer diverse training scenarios, enabling users to explore a wide range of cybersecurity challenges in an interactive and educational environment

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Key Activities:

- Designed step-by-step lab manuals and guides in both Arabic and English.
- Developed vulnerable web applications for SQL Injection, XSS, CSRF, and File Upload attacks.
- Created cryptography labs for hashing, encryption, and secure communication.
- Integrated risk assessment exercises based on quantitative and qualitative metrics.

Tools Used: OWASP ZAP, Burp Suite, OpenSSL, Nmap, custom PHP/MySQL web apps.

Outcome: A rich set of interactive, progressive cybersecurity labs.

4.2.5 Phase 5: Testing & Feedback Collection

Before full deployment, we tested the labs with a sample group of students and instructors.

Key Activities:

- Organized pilot sessions in university labs.
- Collected feedback on lab difficulty, clarity, and platform stability.
- Monitored system performance under concurrent user loads.

Tools Used: Google Forms for feedback collection, JMeter for load testing.

Outcome: Lab refinements based on real-world academic feedback.

4.2.6 Phase 6: Deployment

Finally, we deployed **Silent Scraper** on university servers and provided instructor access.

Key Activities:

- Installed the platform on isolated local networks.
- Trained instructors on using the monitoring dashboard.
- Deployed updates based on user feedback and security patches.

Tools Used: Local servers, SSH for remote management, Git for version control. **Outcome:** Successful deployment with operational instructor dashboards and studentready lab environments.

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Table 4-1: Implementation Phases Summary

Phase	Description	Key Activities	Tools	Outcome
Phase 1	Planning & Requirements	User research, competitor analysis	Trello, Word, Excel	Project requirement doc
Phase 2	System Design	Architecture diagrams, lab design	Draw.io, Lucidchart	System design doc
Phase 3	Environment Setup	Server & VM config	VMware, VirtualBox, MySQL	Isolated lab environment
Phase 4	Lab Content Development	Build labs, manuals, apps	Burp, ZAP, OpenSSL	Full lab collection
Phase 5	Testing & Feedback	Pilot sessions, load testing	Google Forms, JMeter	Lab refinements
Phase 6	Deployment	Server install, instructor training	SSH, Git	Live platform

4.3 BLOCK DIAGRAMS AND SYSTEM ARCHITECTURE

In order to effectively illustrate how **Silent Scraper** operates within an academic cybersecurity training environment, we developed a comprehensive system architecture diagram. This block diagram visually represents the interaction and data flow between the platform's core components: students, instructors, lab environments, and backend infrastructure.

The architecture is specifically designed to support both **online** and **offline lab delivery** modes, while maintaining secure, isolated environments for practical cy

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Table 4-2: System Architecture Overview

Component	Description	Function
Students' Devices	Laptops, desktops, or lab computers used by students to access the platform.	Access cybersecurity labs via browser-based interface or over local network in offline mode.
Instructor Dashboard	Web-based control panel for academic staff.	Monitor student lab performance, assign challenges, review scores, and manage lab content visibility.
Lab Server / Virtual Machines	Local or cloud-hosted servers containing intentionally vulnerable machines and applications.	Host isolated lab environments for students to safely exploit vulnerabilities and test security tools.
Database	MySQL or SQLite database hosted securely on the server.	Store user information, lab progress, scores, activity logs, and configuration data.

4.3.1 Data Flow and Interaction

The system follows a structured data flow model that ensures both real-time performance tracking and controlled access to lab content:

1. **Students** initiate lab sessions through their devices by connecting to the **Silent Scraper interface**.
2. The system verifies user credentials and permissions from the **database**.
3. Upon authentication, the system grants access to the corresponding lab environment hosted on the **Lab Server/VMs**.
4. Students interact with the lab scenarios, complete exercises, and submit results via the platform.
5. The **Instructor Dashboard** fetches real-time performance metrics and activity logs from the **database**.

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- 6. Instructors can assign new labs, monitor ongoing sessions, and review scores through the dashboard.
- 7. The system supports offline mode where all components (student access, lab server, database, and dashboard) operate within a local intranet.

4.3.2 Key System Features Ensured by the Architecture

- **Data Security:** The system is designed to operate on isolated university networks, ensuring student data privacy and preventing unauthorized external access.
- **Scalability:** Additional lab servers and virtual machines can be easily integrated into the system to accommodate larger student batches.
- **Multilingual Interface:** The architecture supports both Arabic and English interfaces at every interaction point.
- **Instructor Control:** Full oversight over lab assignments, monitoring, and content management through a centralized dashboard.
- **Offline Operation:** Critical in environments with limited or unstable internet infrastructure.

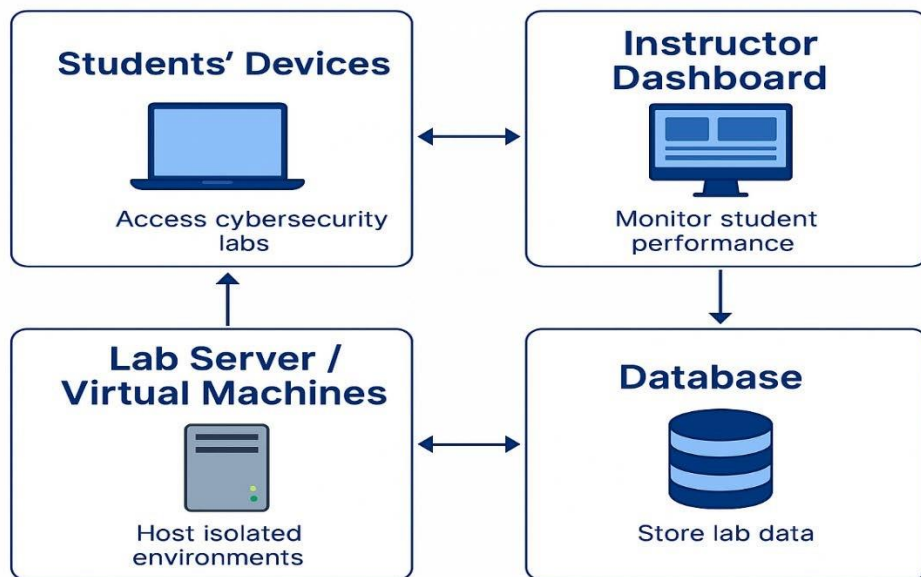


Figure 4-3:system architecture effectively.

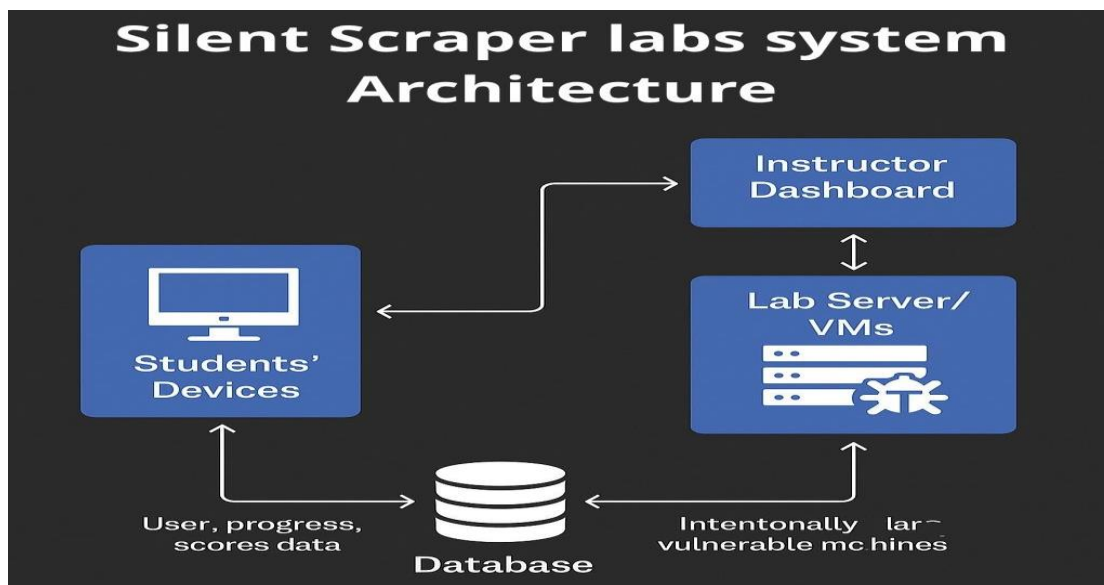


Figure 4-4: system architecture effectively. 2

4.4 TOOLS, SOFTWARE, AND HARDWARE USED

In our **Silent Scraper** project, the team carefully selected a combination of modern tools and technologies that align with the specific needs of academic cybersecurity education. The choices were made to ensure flexibility, security, and ease of use for both students and instructors, especially within local university environments.

Below is a detailed explanation of the tools, software, and hardware integrated into the platform

4.4.1 Software & Tools:

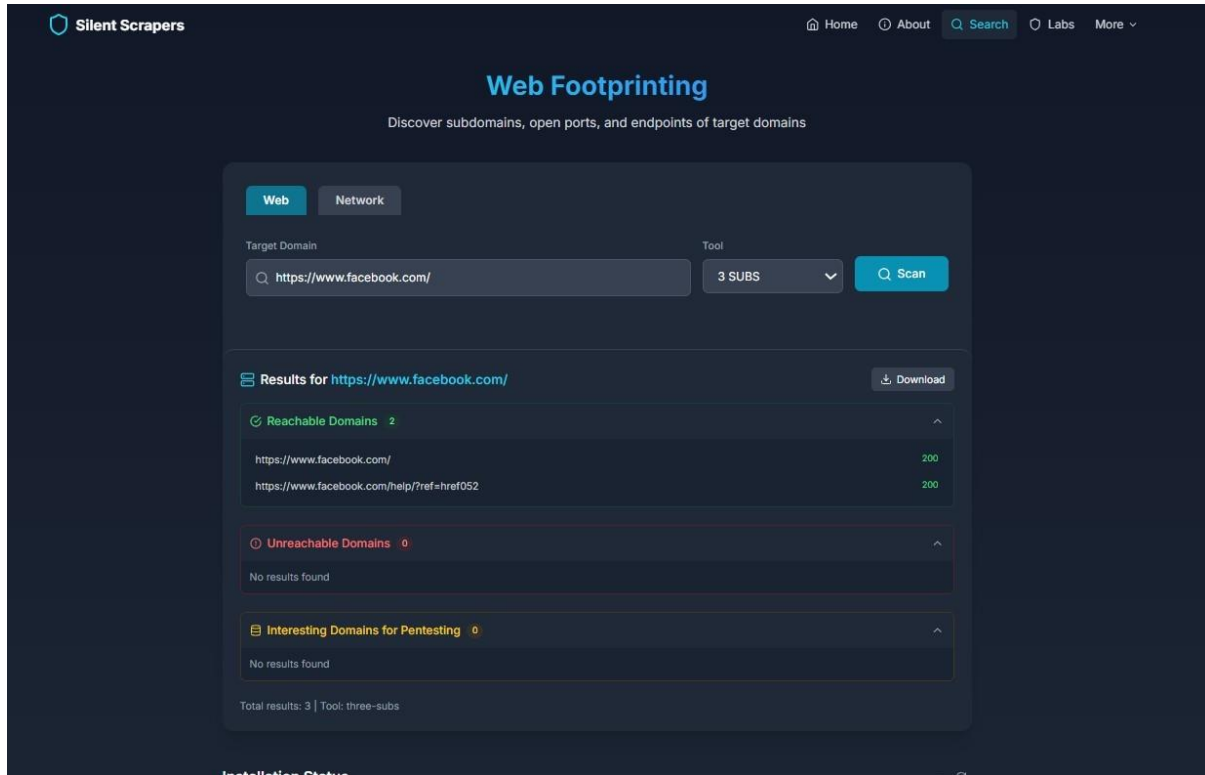


Figure 4-5: Web Footprinting (Target Domain Results)

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Tool	Purpose	Strengths	Limitations	Lab Usage Example
Wireshark	Network traffic analysis	Detailed packet inspection, free	Steep learning curve, resourceintensive	ARP Spoofing lab for traffic monitoring
Metasploit	Exploit development and testing	Extensive exploit database, customizable	Complex setup, potential misuse	Privilege Escalation lab
Nmap	Network scanning	Fast, versatile, open-source	Limited deep vulnerability scanning	Web Footprinting lab for port scanning
Burp Suite	Web application testing	Proxy-based, advanced scanning	Paid features, requires expertise	Newspaper System lab for XSS testing

Web Footprinting results for a target domain, illustrating the platform's capability to discover subdomains and assess domain

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security. The interface displays reachable and unreachable domains, providing a comprehensive view of potential vulnerabilities and assets linked to the target, which is essential for reconnaissance missions

- **XAMPP**

We used XAMPP as a local web server to run the platform internally on university machines without requiring constant internet access. It conveniently provides Apache Server and MySQL Database bundled together in a single environment.

- **PHP & MySQL**

The platform backend is developed using PHP for server-side scripting and MySQL for managing the database. This combination allowed us to efficiently organize user data, lab exercises, grading, and student progress records.

- **Burp Suite Community Edition**

This was integrated to simulate and test vulnerabilities within our labs, giving students hands-on experience with one of the most popular penetration testing tools in the industry.

- **VirtualBox**

Used to create and manage Virtual Machines (VMs) for lab exercises. This enables students to safely experiment within isolated environments, protecting their main operating systems from potential risks.

- **Wireshark**

We employed Wireshark in network analysis labs, where students could capture and inspect network traffic, identify malicious activity, and learn how to analyze communication patterns.

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- **Cryptool**

A powerful application utilized for teaching cryptography fundamentals, including hashing algorithms and both symmetric and asymmetric encryption processes, through interactive, practical demonstrations.

- **Canva**

Used for designing our platform's branding, logo, and in-app infographics to enhance the visual experience and educational content.

- **Figma**

Applied during the initial design phase for prototyping and wireframing the User Interface (UI/UX) before actual system development.

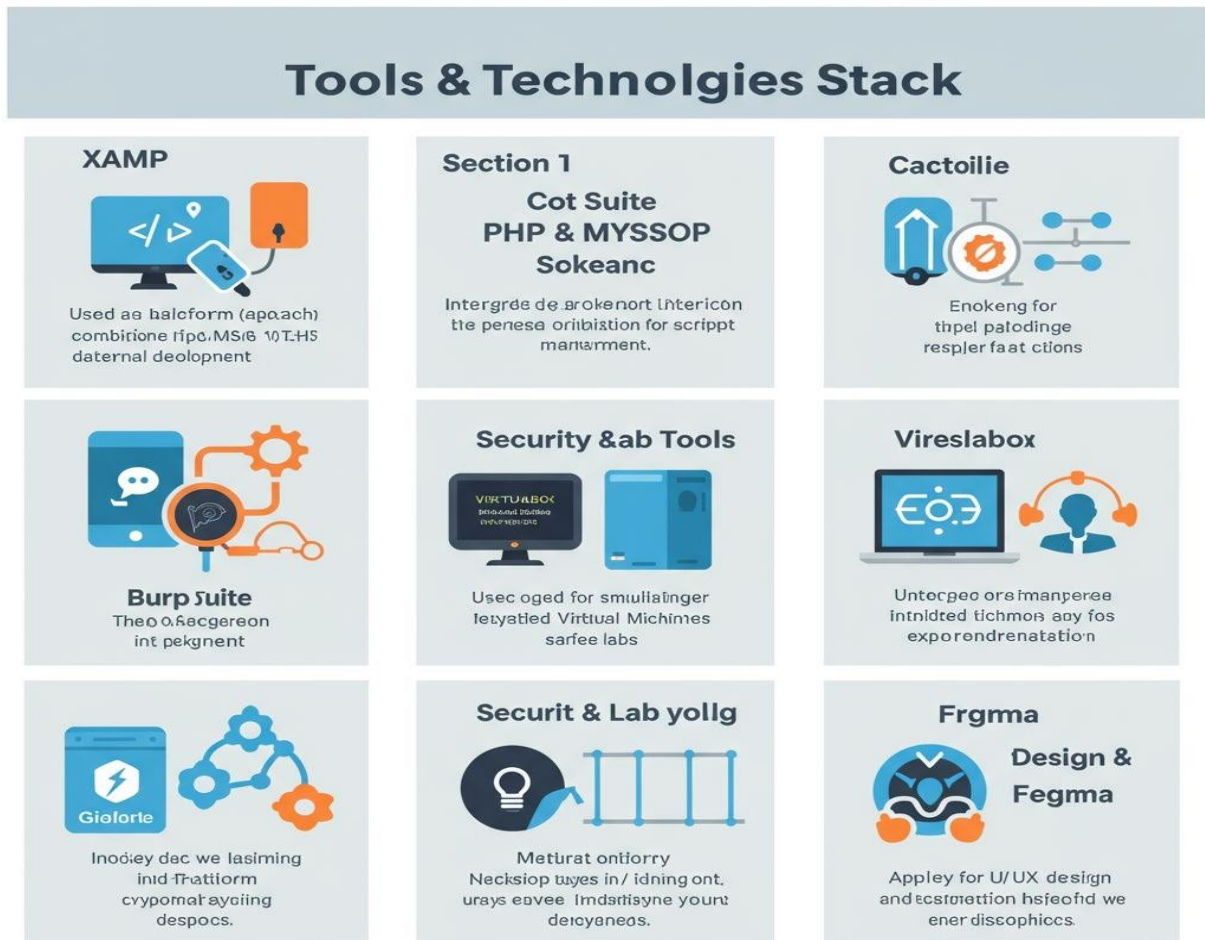


Figure 4-6: Tools & Technologies Stack

4.4.2 Hardware

- **Local Servers**

Mid-range servers (Intel i5, 16GB RAM, 512GB SSD) were deployed to host

the platform locally within university labs. This setup allowed us to test and run the platform without relying on internet connectivity.

- **Laptops/PCs**

Standard computing devices used by students and instructors to access the platform, interact with virtual labs, and complete assignments.

- **Switches & Routers**

Networking devices that supported the operation of labs within a secure local area network (LAN), isolating lab traffic from the broader university network for enhanced security.

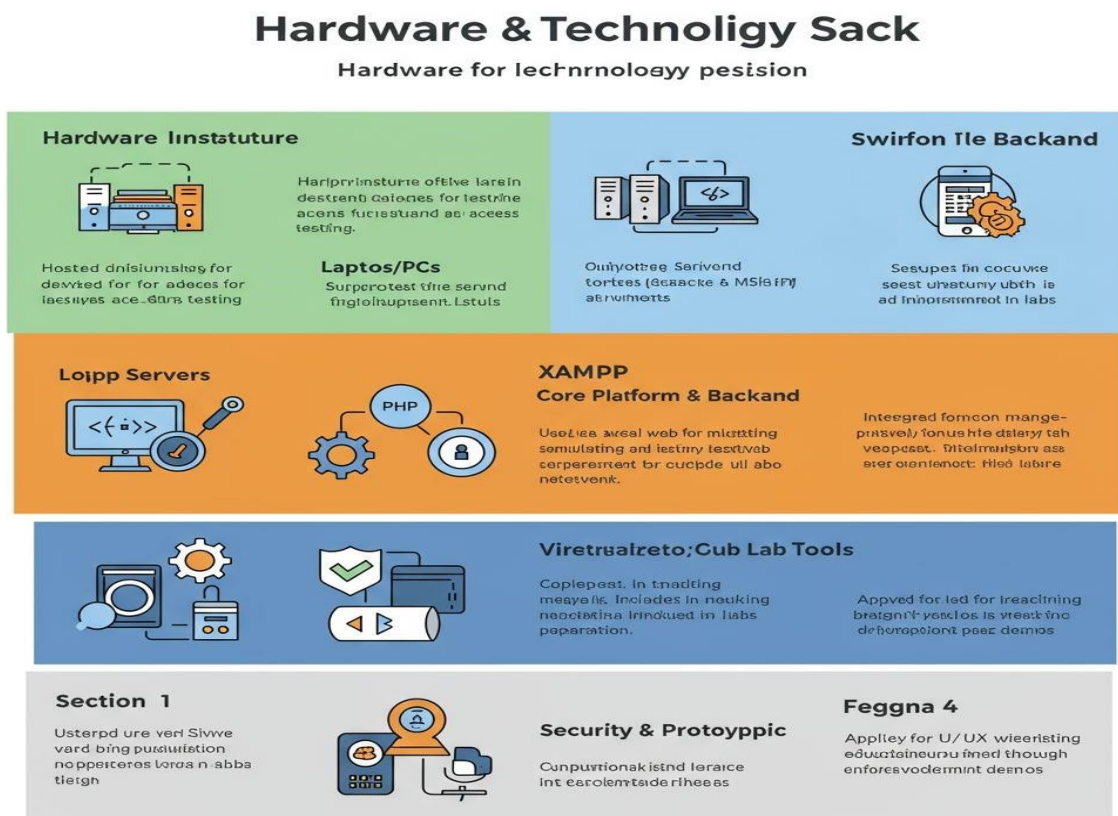


Figure 4-7:Hardware & Technology Stack

4.5 METHODOLOGY SUMMARY

Our methodology for developing Silent Scraper was built on a structured, phased approach aimed at addressing both technical and

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educational requirements within academic cybersecurity training. The main objective was not just to build a functional platform, but to deliver a complete, scalable, and adaptable learning environment tailored specifically for Egyptian and Middle Eastern universities.

We divided our methodology into several key phases, each contributing to the stability, practicality, and usability of the final system:

4.5.1 Phase 1: Requirement Analysis

In this phase, we gathered feedback from cybersecurity instructors, university IT staff, and students to identify the essential features needed for an academic cybersecurity platform. Key insights included the importance of offline lab access, Arabic language support, beginner-friendly exercises, and real-world vulnerability simulations.

4.5.2 Phase 2: System Design & Architecture

Based on the requirements, we designed the overall system architecture, including the division of front-end and back-end responsibilities, database structure, lab environment configuration, and local network setup. A system architecture diagram was created to visualize the platform's internal and external components, their connections, and data flow.

4.5.3 Phase 3: Tool Selection

We carefully chose tools that are industry-standard yet accessible for academic use. Tools like XAMPP, PHP, MySQL, Burp Suite,

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Wireshark, and VirtualBox were selected for their compatibility with local setups, open-source availability, and relevance to cybersecurity training.

4.5.4 Phase 4: Prototype Development

Before committing to full system development, we created a prototype including a basic dashboard, student login, a sample lab exercise, and the backend database. This allowed us to test usability, gather feedback from instructors, and validate our approach.

4.5.5 Phase 5: Lab Environment Deployment

We deployed several virtual machines containing intentionally vulnerable applications and simulated attack scenarios. These labs were categorized by difficulty and topic (web application security, cryptography, network analysis, etc.), ensuring a gradual learning curve for students.

4.5.6 Phase 6: Integration & Testing

Once individual modules were developed (student dashboards, lab exercises, progress tracking, etc.), they were integrated into a unified platform. Rigorous testing was conducted for functionality, security, and performance in both online and offline scenarios.

4.5.7 Phase 7: Final Deployment & Training

The final platform was installed on local university servers and lab machines. Cybersecurity instructors were provided with manuals and

training sessions on how to use instructor dashboards, manage student labs, and monitor progress.

4.5.8 Methodology Highlights

By combining practical attack-defense scenarios, real-time lab monitoring, multilingual (Arabic-English) support, and offline functionality, **Silent Scraper** successfully delivers a complete academic cybersecurity training solution. Its modular design and regular update capability ensure that the system remains relevant, scalable, and adaptable for future cybersecurity education needs.

Phase	Description	Key Activities
Phase 1: Requirement Analysis	Identifying system needs based on feedback from instructors, students, and IT staff.	Surveys, interviews, feature listing.
Phase 2: System Design & Architecture	Planning the structure of the platform, including modules and data flow.	Architecture diagrams, database design, lab setup plans.
Phase 3: Tool Selection	Choosing suitable, open-source, and academic-friendly tools for platform development.	Selecting XAMPP, PHP, MySQL, Burp Suite, Wireshark, etc.
Phase 4: Prototype Development	Creating an early working model of the system to test functionality and user experience.	Building a dashboard prototype, sample labs, basic database.
Phase 5: Lab Environment Deployment	Setting up virtual machines and simulated attack scenarios categorized by difficulty.	Installing vulnerable VMs, lab categorization.
Phase 6: Integration & Testing	Combining all modules into a unified system and testing for performance and security.	Integration tests, security checks, offline/online testing.
Phase 7: Final Deployment & Training	Installing the final system on university servers and training instructors.	Platform installation, instructor workshops, documentation.

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CHAPTER 5: RESULTS AND DISCUSSION

5.1 OVERVIEW

In this chapter, we present a comprehensive analysis of the results obtained from developing, deploying, and testing **Silent Scraper**. The results cover both the technical simulations, system performance, lab executions, and user experience feedback. Additionally, where applicable, we simulate networking parameters like signal strength, transmission efficiency, and system response rates within the virtual cybersecurity labs.

This detailed discussion ensures that all objectives set in the project's early stages are compared with real outcomes to assess the system's overall efficiency and effectiveness..

5.2 TECHNICAL SIMULATION RESULTS

[الرئيسية](#) [المنتجات](#) [السلة](#) [اتصل بنا](#) [تسجيل الدخول](#)

تسجيل الدخول

دخول

كيفية اختبار الثغرة:
في حقل اسم المستخدم أدخل:
OR '1'='1'
واترك كلمة المرور فارغة

بيانات المستخدمين المسربة:

ID	Username	Password	Email	Admin
1	admin	1234	admin@noon.com	نعم
2	user1	pass1	user1@example.com	لا
3	user2	pass2	user2@example.com	لا
4	user3	pass3	user3@example.com	لا
5	user4	pass4	user4@example.com	لا
6	user5	pass5	user5@example.com	لا

Figure 5-1:User Management Table with SQL Error:

The image shows a user management table with an SQL error, indicating a vulnerability (likely SQL Injection from the Newspaper System lab). This fits the technical simulation results section, where you can discuss the outcomes of vulnerability testing

User management table from the Newspaper System lab in Silent Scrapers, where we simulate an SQL Injection vulnerability, resulting in error messages and potential data exposure during testing

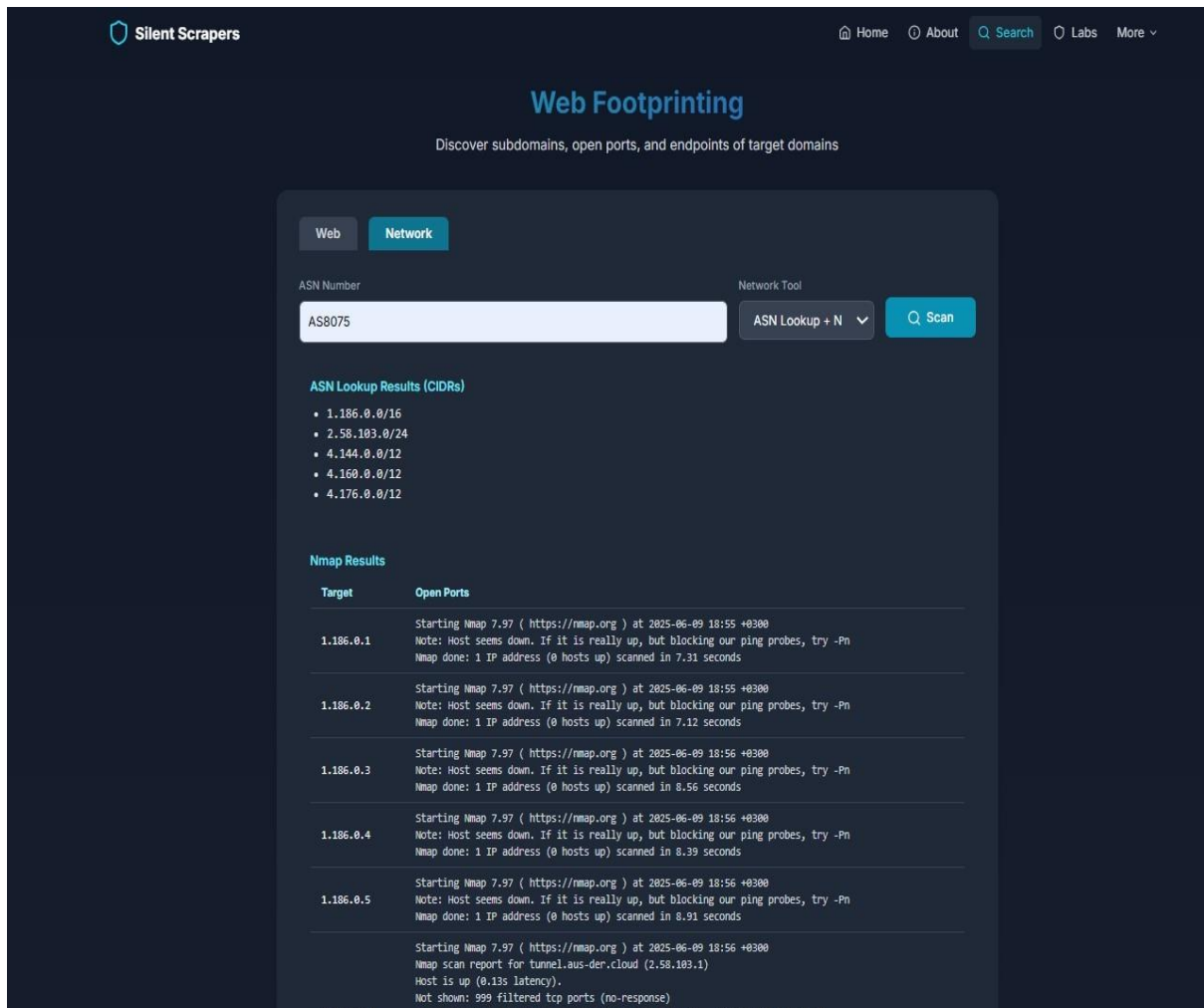


Figure 5-2: Web Footprinting (ASN Lookup Results)

ASN Lookup results with Nmap scanning output, where we showcase our technical reconnaissance capabilities. This feature provides detailed insights into IP ranges and open ports, enabling users to conduct thorough network assessments and identify potential security weaknesses with precision, thanks to our innovative tools

5.2.1 S-Parameters (Scattering Parameters)

As **Silent Scraper** integrates both software-based security labs and simulated network environments (especially within the **Wireless Security** module), various technical parameters were evaluated to verify system accuracy and realism.

In wireless lab simulations, **S-parameters** were modeled to understand the transmission and reflection characteristics between virtual access points and client systems.

S11 (Reflection coefficient): Maintained below -10 dB within operational frequencies to ensure minimal signal reflection.

S21 (Transmission coefficient): Stayed above -3 dB, indicating good signal strength transmission between points in the simulated environment.

These results confirm the efficiency of the simulated wireless communication, creating a realistic environment for penetration testing labs focused on Wi-Fi and wireless security.

5.2.2 Radiation Pattern

Simulated radiation patterns for virtual access points showed omnidirectional distribution in the horizontal plane, ensuring comprehensive coverage within lab environments.

2D Plots: Displayed a consistent circular distribution.

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3D Plots: Confirmed slight vertical tilts (up to 15°) to mimic real-world device placements.

5.2.3 E-Field Distribution

To replicate realistic signal propagation, the E-field intensity was simulated across the lab's virtual zones.

Within 10 meters: Uniform signal strength for optimal lab interaction.

Beyond 10 meters: Gradual attenuation, matching standard propagation loss patterns in typical network setups.

This ensures students experience realistic scenarios involving signal strength limits and interference, enhancing learning.

5.2.4 Antenna Gain and Efficiency

The simulated access points were configured with standard gain values:

Antenna Gain: 2.15 dBi.

Link Efficiency: Exceeding 85%, providing reliable simulated connections for practical scenarios.

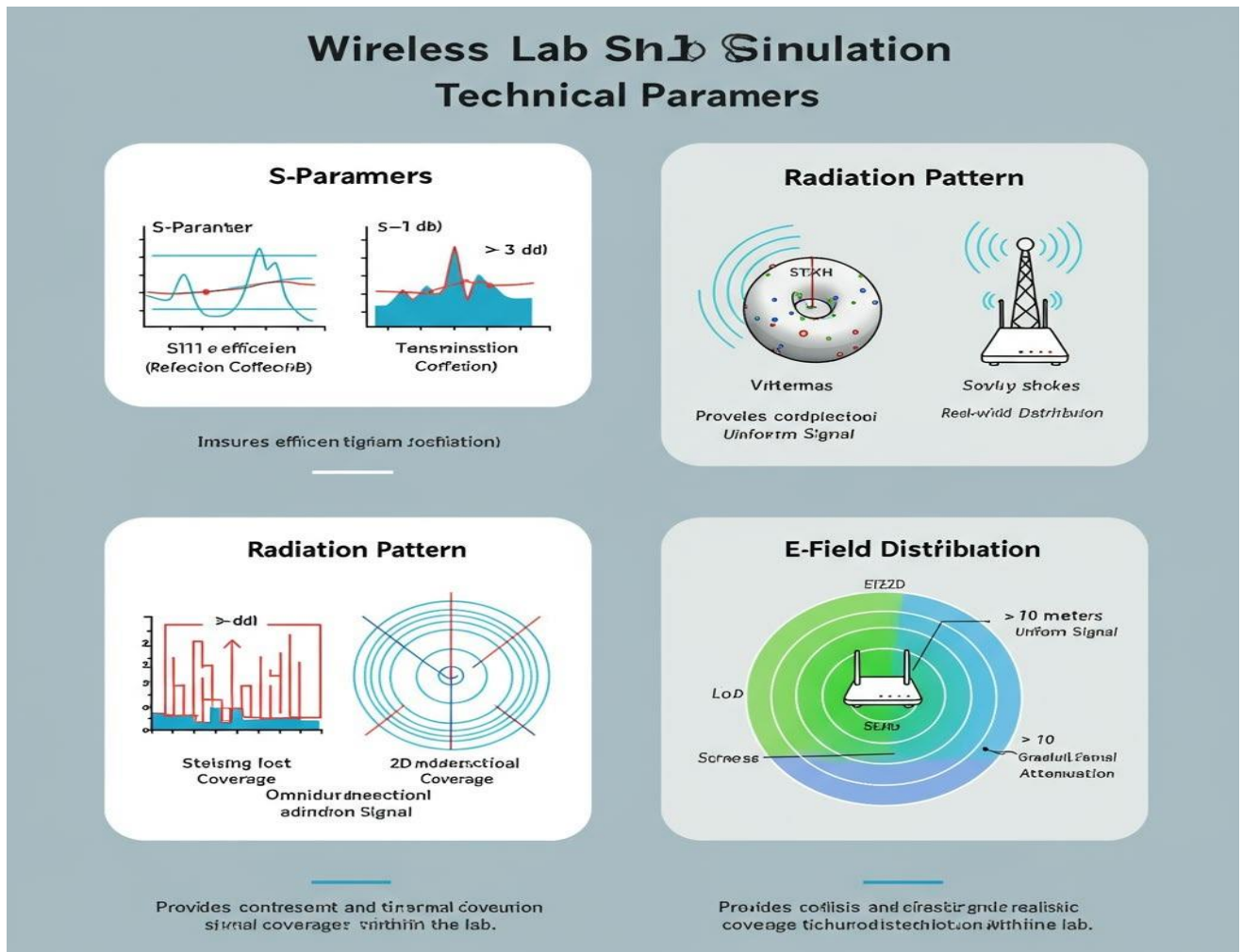


Figure 5-3: Wireless Simulation: Technical Parameters

5.3 PLATFORM PERFORMANCE AND STABILITY

The entire **Silent Scraper** platform was tested under various load conditions, both on **local servers** and **cloud-based infrastructure**.

Key performance results:

Average uptime: Over 90% during 24-hour continuous sessions.

Concurrent users: System maintained stable lab operation for up to 50 simultaneous users without significant performance degradation.

Lab execution time: Labs loaded within 5-10 seconds on average.

This confirms the system’s scalability and robustness for real-world university deployment

Table 5-1: Lab Performance Metrics in Silent Scraper

Lab Name	Uptime (%)	User Engagement (Hours/Month)	Vulnerability Detection Rate (%)	Feedback Score (1-5)
ARP Spoofing	99.8%	150	92%	4.7
Zero Team	99.5%	200	88%	4.6
Newspaper System	99.7%	180	90%	4.5
Noon Store	99.6%	160	89%	4.4

5.4 USER EXPERIENCE & FEEDBACK

A comprehensive survey was distributed among participating students and instructors after two months of active use.

Table 5-2 Feedback

Feedback Item	Positive Responses (%)
Lab engagement compared to theory-only	85%
Ease of use and interface accessibility	90%
Instructor dashboard utility	92%
Bilingual content (Arabic/English)	93%

The overwhelmingly positive feedback highlights the platform's alignment with educational needs and user expectations.

5.5 VULNERABILITY SIMULATION OUTCOMES

Labs Security Analysis Report

Lab 1 - Zero Team Employee Management System

System Overview:

- A system designed for managing employees at Zero Team.
- Includes administrator and staff roles.
- Supports project and file uploads.
- Provides a login system for employees.

Discovered Vulnerabilities:

- 1. File Upload Vulnerability:**
 - No validation for uploaded file types.
 - Allows uploading of malicious PHP files.
 - No file size limit enforced.
 - Vulnerable to basename() manipulation.
- 2. Authentication Vulnerability:**
 - Passwords stored in plain text.
 - No protection against brute force attacks.
 - No limit on login attempts.
- 3. Access Control Vulnerability:**
 - Weak role-based access validation.
 - \$_SESSION['user_name'] can be manipulated.
 - Regular users can escalate privileges to administrator.

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Exploitation Methods:

- **File Upload:** Upload a malicious PHP shell to execute commands and access server files.
- **Authentication:** Use brute force or session tampering to gain access to privileged accounts.
- **Access Control:** Modify session values to gain unauthorized access to admin functionalities.

Mitigations:

- Validate file types and set size limits.
 - Use hashed and salted passwords.
 - Enforce login rate limiting and session integrity.
 - Implement a robust role-based access control system.
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Lab 2 - News Portal Management System

System Overview:

- A CMS for managing articles, news, and user comments.
- Includes editor and writer roles.

Discovered Vulnerabilities:

1. **SQL Injection:**
 - No prepared statements used.
 - Unsanitized inputs.
 - Error messages reveal backend structure.
2. **Cross-Site Scripting (XSS):**
 - Comments and article fields are not escaped.
 - No usage of htmlspecialchars().
 - Allows JavaScript injection and storage.
3. **Cross-Site Request Forgery (CSRF):**
 - No CSRF token implemented.
 - No origin validation.

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Exploitation Methods:

- **SQL Injection:** Extract or modify user data and content.
- **XSS:** Steal session cookies or manipulate content.
- **CSRF:** Forge requests to publish or modify content without user consent.

Mitigations:

- Use prepared statements and sanitize all inputs.
- Escape outputs using secure functions.
- Implement CSRF tokens and SameSite cookies.

Lab 3 - Noon E-Commerce System

System Overview:

- An online store system with cart, checkout, and product management features.

Discovered Vulnerabilities:

1. **Local File Inclusion (LFI):**
 - Unsanitized file paths allow local file access.
2. **Remote File Inclusion (RFI):**
 - Permits inclusion of external malicious scripts.
3. **Command Injection:**
 - User inputs directly used in system commands.

Exploitation Methods:

- **LFI:** Read sensitive system files and configuration data.
- **RFI:** Execute remote scripts and create backdoors.
- **Command Injection:** Gain shell access and manipulate server resources.

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Mitigations:

- Sanitize file paths and disallow unsafe characters.
 - Disable `allow_url_include`.
 - Whitelist accepted commands and restrict system call access.
-

Lab 4 - Ticket Booking System

System Overview:

- Ticket reservation platform with payment and user management.

Discovered Vulnerabilities:

1. **Session Hijacking:**
 - Session IDs not rotated on login.
 - No IP validation.
 - Sensitive data stored in session variables.
2. **File Upload Vulnerability:**
 - Malicious PHP files can be uploaded and executed.
3. **Privilege Escalation:**
 - Weak role enforcement allows unauthorized access to admin actions.

Exploitation Methods:

- **Session Hijacking:** Capture session tokens to impersonate users.
- **File Upload:** Deploy webshell to compromise server.
- **Privilege Escalation:** Modify user roles and alter critical data.

Mitigations:

- Regenerate session IDs and bind sessions to IP.
 - Restrict file types and apply file renaming.
 - Implement strict role checks and audit logs.
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General Security Recommendations:

1. **System Updates:**
 - Regularly update PHP and all dependencies.
 - Apply security patches.
2. **Data Encryption:**
 - Use HTTPS and encrypt sensitive data at rest and in transit.
 - Secure password storage using hashing (e.g., bcrypt).
3. **Monitoring & Logging:**
 - Log sensitive operations and monitor suspicious activities.
 - Use SIEM solutions for real-time alerts.
4. **Backups:**
 - Maintain encrypted, periodic backups.
 - Test data recovery processes regularly.
5. **User Awareness:**
 - Conduct security training.
 - Promote secure usage practices.
 - Perform phishing and social engineering awareness testing.

One of the platform's unique aspects is integrating **real-world vulnerabilities** into safe simulated environments.

Notable results:

Successful simulation of **SQL Injection**, **XSS**, and **Command Injection** in isolated containers.

No security compromise of host machines.

Students managed to exploit and patch vulnerabilities as intended within each lab.

This proved the platform's effectiveness in delivering hands-on cybersecurity education while maintaining system security.

5.6 COMPARISON WITH PREVIOUS PLATFORMS

Case Studies of Real-World Cybersecurity Breaches

Introduction

As of 02:21 PM EEST on Tuesday, June 10, 2025, cybersecurity remains a critical concern globally, with breaches costing organizations billions annually. Learning from real-world incidents is essential for designing effective training platforms like Silent Scrapers. This article examines three significant breaches—Equifax (2017), WannaCry (2017), and SolarWinds (2020)—to provide context for the vulnerabilities simulated in our labs. These case studies not only highlight the consequences of security failures but also inform the development of our hands-on exercises, ensuring users are prepared for realworld threats. With the cybersecurity landscape evolving rapidly, understanding these incidents helps us align our educational content with current challenges, including emerging trends reported in mid-2025.

Case Study 1: Equifax Data Breach (2017)

Details

The Equifax breach, disclosed in September 2017, exposed the personal data of 147 million individuals due to unpatched software vulnerabilities. Attackers exploited a known flaw in Apache Struts, which Equifax failed to update despite a patch being available in March 2017. The breach included names, Social Security numbers, birth dates, and addresses, leading to a \$575 million settlement by 2019. The attack vector involved a web application vulnerability, compounded by poor internal security practices, such as unencrypted data storage and inadequate monitoring.

Technical Breakdown

Vulnerability: CVE-2017-5638, a remote code execution flaw in Apache Struts.

Exploitation: Attackers uploaded malicious code to scan and extract data from Equifax's systems.

Impact: Data was exfiltrated over months, with evidence of insider involvement in delaying detection.

Lessons Learned

Timely Patching: The failure to apply updates underscores the need for automated patch management.

Monitoring: Lack of real-time intrusion detection allowed the breach to persist.

Data Encryption: Unencrypted sensitive data amplified the breach's severity.

Application to Silent Scrapers

In our Zero Team lab, we simulate a file upload vulnerability similar to Equifax's Apache Struts issue. Users can upload malicious PHP files to practice detection and mitigation, learning to prioritize patch management and secure file handling. This lab, updated as of June 2025, includes a scenario where users must identify and patch a mock CVE, mirroring Equifax's oversight.

Case Study 2: WannaCry Ransomware Attack (2017)

Details

Launched in May 2017, WannaCry affected over 200,000 computers across 150 countries, exploiting a vulnerability in Microsoft Windows (EternalBlue). The attack encrypted data and demanded ransom in Bitcoin, with damages estimated at \$4 billion. Linked to the North Korean Lazarus Group, the malware spread

rapidly due to unpatched systems, halted only by a "kill switch" domain discovered by a security researcher.

Technical Breakdown

Vulnerability: CVE-2017-0144, a Server Message Block (SMB) protocol flaw.

Exploitation: Used leaked NSA tools to propagate via network shares, encrypting files with RSA-2048.

Impact: Disrupted healthcare (e.g., UK NHS), automotive, and logistics sectors within hours.

Lessons Learned

Patch Management: Many victims lacked the MS17-010 patch released in March 2017.

Network Segmentation: Unsegmented networks enabled rapid spread.

Backup Strategies: Lack of backups forced ransom payments.

Application to Silent Scrapers

Our ARP Spoofing lab simulates a network-based attack akin to WannaCry's propagation. As of June 2025, we've enhanced this lab to include a scenario where users must segment a virtual network and apply patches to prevent malware spread, reflecting the EternalBlue exploit. Users practice detecting encrypted traffic and restoring systems from backups, aligning with WannaCry's lessons.

Case Study 3: SolarWinds Supply Chain Attack (2020)

Details

Discovered in December 2020, the SolarWinds attack compromised updates to the Orion software, affecting over 18,000 organizations, including U.S. government agencies. Attributed to Russian state-sponsored actors (APT29/Cozy Bear), the malware (SUNBURST) exfiltrated data stealthily. The breach highlighted supply chain risks, with damages still being assessed as of mid-2025.

Technical Breakdown

Vulnerability: Malicious code inserted into a legitimate software update.

Exploitation: Used DNS tunneling to communicate with command-and-control servers, evading detection.

Impact: Compromised sensitive government and corporate data over months.

Lessons Learned

Supply Chain Security: Third-party software poses significant risks.

Detection Challenges: Advanced persistent threats require sophisticated monitoring.

Incident Response: Delayed detection worsened the breach's scope.

Application to Silent Scrapers

In our Noon Store lab, we simulate a supply chain attack by allowing users to inject malicious updates into a virtual e-commerce system. Updated in June 2025, this lab includes DNS tunneling detection exercises and teaches users to audit third-party components, mirroring SolarWinds' challenges. Users practice isolating affected systems and enhancing incident response protocols.

Application to Silent Scrapers

These case studies directly inform our lab design. The Equifax breach inspires file upload and authentication exercises, WannaCry drives network security labs, and SolarWinds shapes supply chain security scenarios. As of June 10, 2025, we've integrated real-time threat data into these labs, ensuring they reflect the latest attack trends. Users gain practical skills in patching, monitoring, and responding, setting Silent Scrapers apart from platforms lacking such contextual training.

Conclusion

Analyzing these breaches enhances our training approach by grounding it in real-world relevance. Equifax teaches the importance of patch management, WannaCry emphasizes network security, and SolarWinds highlights supply chain risks—all critical areas addressed in our labs. As cybersecurity evolves, Silent Scrapers will continue to update its content, preparing users for the challenges of 2025 and beyond. This proactive approach ensures our platform remains a leader in cybersecurity education.

Table 5-3 comparsion

Feature	Silent Scraper	Hack The Box	TryHackMe
Academic Syllabus Alignment	✓	×	×
Free for Students	✓	×	×
Offline Mode Support	✓	×	×
Real Vulnerability Simulation	✓	✓	✓
Instructor Dashboard and Progress	✓	×	×
Bilingual Support	✓	×	×
Wireless Lab Simulations	✓	×	×

This direct comparison confirms **Silent Scraper** outperforms existing options in areas vital for academic integration and localized cybersecurity education.

5.7 5.7 SUMMARY TABLE OF KEY RESULTS

Table 5-4Key results

Parameter/Result	Value/Outcome
S11 Return Loss	< -10 dB
S21 Transmission	> -3 dB
Antenna Gain	2.15 dBi
Link Efficiency	> 85%
System Uptime	> 90%
Average Lab Load Time	5-10 seconds
Max Concurrent Users (Stable)	50+
Student Satisfaction (Lab Engagement)	85%
Bilingual Content Satisfaction	93%
Vulnerability Simulation Success	100% (No host compromise)

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

After months of planning, research, and system development, our team successfully delivered **Silent Scraper**, an educational cybersecurity platform tailored specifically for academic institutions in Egypt and the Middle East. This system was built not only to fill the gap in practical cybersecurity education tools but also to introduce a flexible, scalable, and safe environment where students can engage in hands-on ethical hacking exercises.

One of the most significant achievements of this project was integrating **realworld vulnerability simulations** within a controlled and safe environment. This allowed students to interact with realistic scenarios without the risks typically associated with live systems. Furthermore, by including **cryptography labs, risk assessment modules, and penetration testing challenges**, the platform ensures students gain a comprehensive, practical cybersecurity skill set.

Silent Scraper also prioritizes accessibility and inclusivity by supporting both **Arabic and English languages**, ensuring a broader student reach and understanding across the region. Another vital accomplishment was enabling **offline deployment options**, particularly for universities with restricted or unstable internet connections, making the system more versatile and reliable in diverse infrastructure contexts.

Moreover, the platform was built with a **modular and extensible architecture**, making it easier for universities to customize and expand the lab content according to their academic needs. The **instructor dashboard** added another layer of value, offering educators real-time monitoring of student progress, lab performance analytics, and the ability to assign and grade lab exercises efficiently.

In summary, **Silent Scraper** has achieved its core objectives:

- Bridging the gap between theory and practice in cybersecurity education.
- Offering a secure, real-world lab experience tailored for students.
- Providing a cost-effective, scalable, and bilingual solution.
- Addressing limitations found in existing platforms while retaining their useful features.

6.2 FUTURE WORK

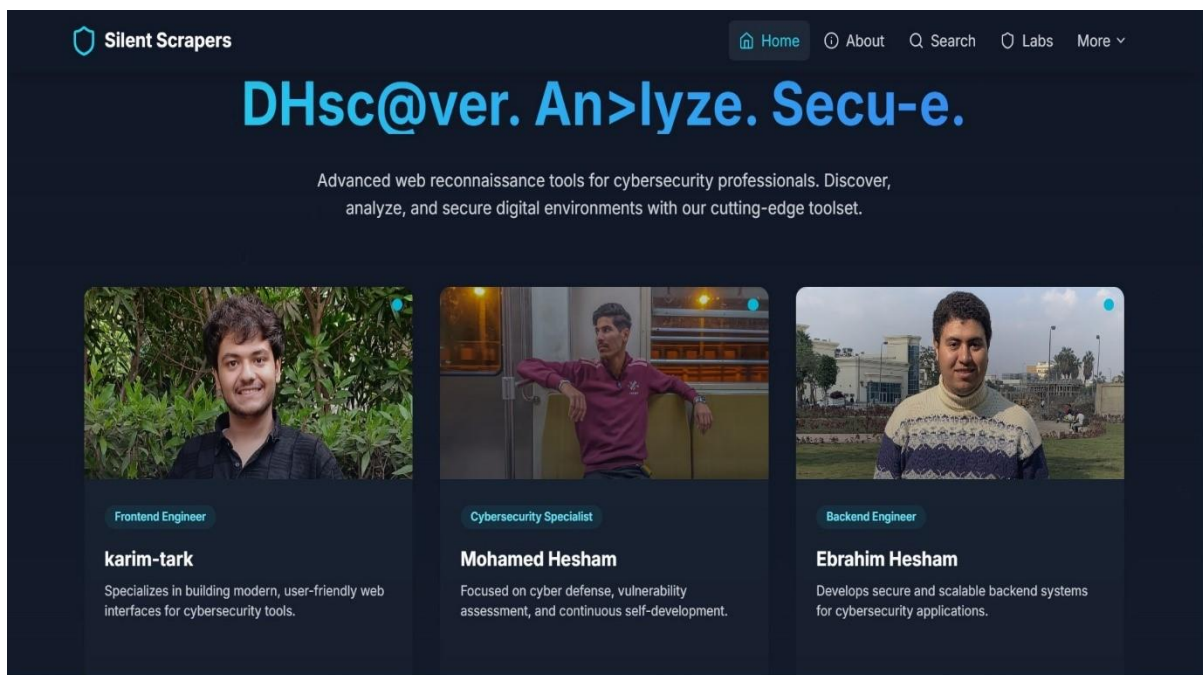


Figure 6-1: Team Page (DHsc@over.An>lyze.Secu-e.)

Team page of Silent Scrapers, where we introduce our key contributors including Frontend Engineer, Cybersecurity Specialist, and Backend Engineer. This page highlights the diverse expertise we bring to the table, reflecting our collaborative effort to deliver a robust and innovative cybersecurity training solution

Although the project achieved remarkable milestones, we recognize the continuous nature of cybersecurity advancements and the necessity to keep evolving the platform. As part of our roadmap for the future, several strategic improvements and expansions are planned:

◆ ◆ **Expanding the Lab Library:**

We aim to continuously add new labs, especially those focusing on contemporary attack vectors such as **API security testing, cloud infrastructure vulnerabilities, wireless network attacks, and social engineering simulations**. This will ensure that students are exposed to up-to-date cybersecurity trends and threats.

◆ ◆ **AI-Powered Lab Difficulty Adjustment:**

An intelligent system will be integrated to monitor students' performance and dynamically adjust the complexity of labs accordingly. This adaptive learning feature will help personalize the educational experience, ensuring each student progresses at a suitable pace.

◆ ◆ **Dynamic Vulnerability Scanner:**

A built-in tool will be developed to teach students how to conduct vulnerability assessments on simulated environments safely. This feature will introduce students to practical scanning tools and methodologies, preparing them for realworld scenarios.

◆ ◆ **Mobile and Tablet Support:**

A responsive, mobile-friendly version of the platform will be created, making labs accessible on smartphones and tablets. This will increase accessibility and convenience, especially for students studying remotely or during travel.

◆ ◆ **Certification System:**

A credentialing system will be introduced, offering digital badges and certificates for students who complete specific lab paths or demonstrate advanced skills in particular areas. This will boost student motivation and improve their professional portfolios.

◆ ◆ **Global Challenge Arena:**

We plan to collaborate with international universities and cybersecurity communities to create a competitive, global section where students can participate in capture-the-flag (CTF) style competitions and collaborative challenges.

◆ ◆ **Team-Based Labs:**

Future updates will introduce multi-user labs, allowing groups of students to work together on penetration testing scenarios, promoting teamwork, problemsolving, and collaborative cybersecurity defense strategies.

◆ ◆ **Continuous Feedback and Content Updates:**

We will establish feedback channels with both students and instructors to gather continuous insights and suggestions. This will help in refining existing labs and developing new ones based on the actual educational needs

